Conductor reductions and finite verification for Wilf's inequality

9 October 2026

Abstract

Let S be a numerical semigroup with conductor c, embedding dimension e and $n = |S \cap [0, c)|$. For $S \neq \mathbb{N}$, let m be its multiplicity, $q = \lceil c/m \rceil$ and $t_L = |\operatorname{Ap}(S, m) \cap (0, c)|$. We derive $ne - c \geq (n - q)(e - t_L - 1)$. Every strict Wilf violation has $n \geq 5$ and $c \leq (n - 3)(n + 1)$, giving a finite conductor reduction for each bounded interval of left cardinalities. An ordered fixed-conductor algorithm enumerates exactly the semigroups with negative Eliahou number in any prescribed finite conductor and left-cardinality range. We prove its pruning rules, termination and uniqueness, and give an exhaustive-forest certificate whose checker reconstructs every required branch. Explicit representation and bit-cost bounds yield a deterministic $2^{O(v \log(v+1))}$ upper bound for deciding Wilf's inequality throughout $u \leq n \leq v$. This is a general algorithmic statement, without a new execution or a feasibility assertion. The scalar conductor estimate is a coarse consequence of sharper Apéry-profile and compression bounds. We develop those bounds through selected factorizations, Hilbert functions and joint matchings of values and residues. Their recorded application proves Wilf's inequality for $13 \leq n \leq 37$, using Bacher's reported verification for $c \leq 200$ as an external computational input. A separate mixed-interval family at depth six has strict positivity without a cardinality bound.

1 Introduction

Let $S \subseteq \mathbb{N}$ be a numerical semigroup, with least conductor c and minimal generating set P. Write

$$L = S \cap [0, c), \qquad n = |L|, \qquad e = |P|.$$

Wilf's conjecture [13] asks whether

$$W(S) := ne - c \geq 0$$

for every such semigroup. The left cardinality n counts zero when $c > 0$, and e counts *all* primitives, including those at or above the conductor. For $S = \mathbb{N}$ one has $c = n = 0$, $e = 1$ and $W = 0$.

This paper develops conductor reductions from Apéry profiles and uses them to organize finite verification of Wilf's inequality. Its first conclusion has no prescribed range of left cardinalities. For $S \neq \mathbb{N}$, let m be its multiplicity, put $q = \lceil c/m \rceil$, and let t_L be the number of positive Apéry elements below c. Theorem 1 proves

$$W(S) \geq (n - q)(e - t_L - 1). \tag{1}$$

Consequently $e \geq t_L + 1$ is sufficient for Wilf's inequality. Keeping the strict integer endpoints and excluding depths at most two yields Theorem 2: every strict violation satisfies

$$n \geq 5, \qquad 3 \leq q \leq n - 2, \qquad 2n \leq c \leq (n - q)(n + 1) \leq (n - 3)(n + 1). \tag{2}$$

Thus a fixed interval of left cardinalities reduces to a finite conductor domain. Corollary 9 states this equivalence precisely, including its small-cardinality boundary. The equivalence does not assert that the remaining finite verification has been carried out.

The argument selects one factor pair for each distinct decomposable Apéry value. Most selected pairs carry weight at least q; pairs of weight $q-1$ produce compressed values and share a single allowance of $\rho = qm - c$ residues. Summing the weights of all unordered pairs gives (1). The Apéry and compression ingredients are established tools [7, Lemma 2.13 and Propositions 2.18, 2.20]. The closed-form bound is a coarse consequence of those ingredients: Proposition 10 shows that the more detailed profile bound used later is at least as strong. We make no claim of priority for this scalar consequence or of additional computed coverage from its derivation.

1.1 Exact verification and resource bounds

Theorem 3 gives a uniform algorithm $\textsc{Primary}(C, u, v)$ for all semigroups with $c \le C$, $u \le n \le v$ and negative Eliahou number W_0. Its product and large-generator pruning principles adapt Bacher's arguments [1, §2.2, Lemmas 2.1 and 2.3] to fixed actual conductor and an increasing list of left primitives. We prove the precise transition rules and explain which statements apply only to proper descendants. A full bit-set representation and immutable parent states make the paper-to-pseudocode correspondence explicit.

Theorem 4 supplies an exhaustive-forest certificate: the checker derives the required roots and children and reconstructs all invariants. Valid output records alone do not prove coverage. An accepted complete forest, together with $W \ge 0$ on every W_0-negative output, proves Wilf's inequality in the specified finite region. Combining this with the conductor reduction gives the certificate equivalence in Corollary 17.

The resource analysis distinguishes arithmetic on known invariants, extraction from sparse or generator input, and verification of an entire family. Corollary 21 gives a uniform deterministic $2^{O(v\log(v+1))}$ upper bound for the bounded-left-cardinality decision problem. This upper bound is not polynomial in the binary length of v; it provides no lower bound or claim that the remaining region has been checked. Detailed residual-domain counts explain how depth, primitive width, monomial degree and box widths affect the profile method. Classical residue-graph methods are credited in Section 6.2.

1.2 Profiles, selected factorizations and applications

The sharper reduction retains the full Apéry profile. We use a divisor-closed selection of Apéry factorizations, following Eliahou's divset construction [6, Proposition 2.15]. We select one factorization of each represented Apéry value by minimizing degree and then an additive monomial rank. The resulting standard graded quotient allows Macaulay's theorem to bound higher-degree values using their divisors. Hilbert functions have previously been used to estimate iterated sumsets by Eliahou and Mazumdar [8, 9]. Here the selected quotient restricts the possible profiles, and exact matchings of integer values and residues exclude the remaining primitive-value boxes.

This construction also has a uniform structural application. Theorem 6 proves strict positivity of the Eliahou number for a specified mixed-interval family at depth six, without a bound on the number of left elements. Its hypotheses concern the exact positions of the left primitives; they are not consequences of a small cardinality assumption. The same selected order ideal therefore supports both a structural argument and the finite profile exclusions.

The recorded computational application is Theorem 7: Wilf's inequality holds for $13 \le n \le 37$. Eliahou and Marín-Aragón established the range $n \le 12$ [7, Theorem 4.1]; an earlier proof of $n = 13$

appears in [12, Theorem 12.1]. The application retained here combines the profile and matching arguments with Bacher's reported verification for $c \le 200$ [1, §2.3]. The reduction at $n = 13$ is algebraic; the recorded residual traversal covers $14 \le n \le 37$. Section 11 gives the exact scope and validation limits of that computation. Neither the scalar conductor bound nor its presentation here replaces this external conductor input or constitutes a new computational run.

1.3 Notation and conventions

Except when $S = \mathbb{N}$ is handled explicitly, put

$$m = \min(S \setminus \{0\}), \quad q = \lceil c/m \rceil, \quad \rho = qm - c, \quad A = \mathrm{Ap}(S, m) = \{a \in S : a - m \notin S\}.$$

Here $c \ge 2$, $m \ge 2$ and $0 \le \rho < m$. The shifted level and layer are

$$\lambda(x) = \left\lfloor \frac{x + \rho}{m} \right\rfloor, \qquad A_i = \{a \in A : \lambda(a) = i\}.$$

Let D denote the *finite* set of positive decomposable elements of A, and put $\delta = |D|$. In particular, D is not the infinite set of all decomposable elements of S. Set

$$T = A \cap (0, c), \quad t_L = |T|, \quad \nu = n - q, \quad R = T \cap D, \quad B = T \setminus D, \quad R_i = R \cap A_i.$$

The symbol t_L denotes a left Apéry cardinality, not the type of a numerical semigroup. For $1 \le i < q$ write

$$\alpha_i = |A_i|, \quad \kappa_i = |R_i|, \quad \beta_i = \alpha_i - \kappa_i, \qquad K = \sum_i \kappa_i, \quad s = \sum_i \beta_i = |B|.$$

We reserve ν for $n - q$; the capital N in some later cap formulas is a separately stated upper bound on a number of noncompressed values.

For the primitive counts, write

$$\ell_P = |P \cap [0, c)|, \qquad r_P = |P \cap [c, c + m)|.$$

Every primitive is less than $c + m$, since any element at least $c + m$ decomposes as $m + (x - m)$. Thus $e = \ell_P + r_P$. The Eliahou number is

$$W_0(S) = n\ell_P + qr_P - c = n\ell_P - q(m - r_P) + \rho. \tag{3}$$

This agrees with the published convention [4, Definition 3.10]. The q multiples $0, m, \dots, (q-1)m$ lie below c, so $n \ge q$ and

$$W(S) - W_0(S) = (n - q)r_P \ge 0. \tag{4}$$

The capital E used in the finite profile reduction is an *embedding-dimension cap*, not the Eliahou number. In particular, an assertion about $W < 0$ must not be silently replaced by an assertion about $W_0 < 0$.

All finite intervals $[a, b]_{\mathbb{Z}}$ are inclusive integer intervals; $[a, b)$ has last integer $b - 1$. Empty sums and products are 0 and 1. Monomials, pair counts and divisor conditions allow repeated variables. We use $\binom{a}{b} = 0$ when $0 \le a < b$. Auxiliary indices, factorization degrees and locally defined counts retain their stated local scopes.

1.4 Inputs used for the recorded application

The elementary conductor reduction in Section 3 does not require the following stronger Wilf criteria or any computational input. They enter the sharper profile application as indicated.

1. Wilf's inequality holds if $e \leq 3$ [10].
2. It holds if $q \leq 3$ [4, Proposition 3.13 and Corollary 6.5]; the case $q = 1$ is the ordinary semigroup $\{0\} \cup [m, \infty)$.
3. If $q \geq 4$ and $\alpha_i = 0$ for $1 \leq i < \lceil q/2 \rceil$, then $W \geq 0$ [7, Theorem 3.1 and Proposition 2.7].
4. If $e, q \geq 4$ and $n \leq 3q - 1$, then $W \geq 0$. This follows from the argument of [5, §5], with the details supplied in Proposition 44.
5. The Hilbert function of a standard graded commutative $\mathbb{Q}$-algebra satisfies Macaulay's growth inequality [4, Theorem 5.5]. The binomial convention is fixed in Lemma 22.
6. Bacher reports verification of Wilf's inequality for every numerical semigroup with $c \leq 200$ [1, §2.3]. This is the external computational input used by the retained application.

1.5 Organization and logical dependencies

Section 2 supplies the Apéry identities and profile bounds; Section 3 derives the general conductor reduction. Sections 4 and 5 give the ordered enumeration and exhaustive checker. Section 6 proves their representation and resource bounds. The detailed profile, shadow and matching methods follow in Sections 7, 8 and 9, with the depth-six companion in Section 10. Section 11 gives the recorded cardinality application. Appendix B specifies its implemented procedure, and Appendix D bounds the mathematical residual domains.

Conclusion	Principal dependencies
General conductor reduction	Apéry identities, one global compression allowance and the elementary depth-two argument.
Finite verification and decision bound	Ordered primitive prefixes, proved descendant gates, exact forest reconstruction and the conductor cutoff.
Depth-six positivity	The selected order ideal and the exact mixed-interval hypotheses; no cardinality bound or recorded traversal.
The 13–37 application	Stronger Wilf criteria, profile/shadow/matching bounds, Bacher's conductor input and the recorded complete residual cover.

The general enumeration and checker are proved specifications. They have not been implemented or executed as part of the results reported here. The recorded application retains its original rules and counts; the new scalar reduction and general algorithm are not additional gates silently inserted into that historical traversal. Nor does a proof of the present algorithm retrospectively validate a different program or its reported run.

2 Apéry identities and profile bounds

Lemma 1 (Apéry partition and universal subsums)**.** *There is one element of A in each residue modulo m, $A \subset [0, c+m)$, $A_0 = \{0\}$, and every positive element of A has level between 1 and q. Every positive partial sum in* any *representation of an Apéry element as positive semigroup elements is Apéry, and every proper such partial sum is less than c. In particular every element of D is a sum of members of B. Moreover,*

$$m = e + \delta, \qquad n = q + \sum_{i=1}^{q-1} (q-i)\alpha_i, \tag{5}$$

$$c = q(e+\delta) - \rho, \qquad W < 0 \iff q\delta - \rho > (n-q)e. \tag{6}$$

We have $1 \le q \le n$, $\kappa_1 = 0$, and $e \ge \max(2, 1+s)$.

Proof. The Apéry representative of a residue is its least element in S; every element of S is uniquely $a + tm$ with $a \in A$ and $t \ge 0$. Since $a - m \in S$ for $a \ge c + m$, we have $A \subset [0, c+m)$. A positive Apéry element is at least $m+1$, giving its level range. Every primitive other than m lies in A, since $p = m + (p-m)$ would otherwise decompose it. Partitioning A into zero, these $e-1$ primitives, and D gives $m = e + \delta$; in particular e includes right primitives.

If $a = x_1 + \cdots + x_t \in A$ and a positive partial sum v satisfied $v - m \in S$, then $a - m \in S$, a contradiction. A proper partial sum has a complement of at least m, so $v \le a - m < c$. Decomposing positive elements into primitives shows that every $a \in D$ is a sum of members of B. Each positive summand raises the level, so no level-one Apéry element is decomposable.

Write $a + \rho = im + r$, $0 \le r < m$. The progression $a + tm$ contributes $q - i$ elements below c if $i < q$, and none if $i = q$. The zero progression contributes q, proving (5) and $q \le n$; substitution gives (6). The primitives m and B give $e \ge 1 + s$. Since a numerical semigroup with only one primitive is $\mathbb{N}$, we also have $e \ge 2$ here. □

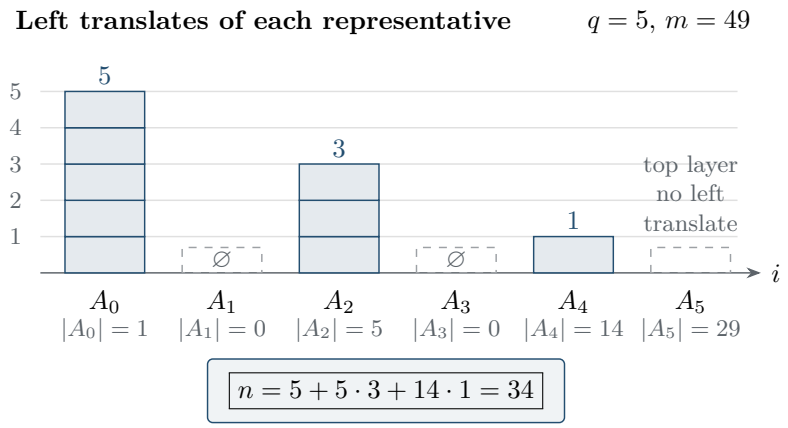


Figure 1. The Apéry staircase for the executed relaxed profile $q = 5$, $\alpha = (0, 5, 0, 14)$ at $n = 34$, displayed with the exact tuple $m = 49$. The ∅ symbols mark empty layers. Each representative in A_i contributes $q - i$ elements below c by adding multiples of m; $0 \in A_0$ contributes five. Thus $n = 5 + 5 \cdot 3 + 14 \cdot 1 = 34$. The top layer contributes zero. The profile and tuple belong to the necessary-case enumeration.

Lemma 2 (Level bounds and the empty boundary)**.** *For positive $x, y \in S$,*

$$\begin{gathered}\lambda(x)+\lambda(y)-1 \le \lambda(x+y) \le \lambda(x)+\lambda(y)+1,\\ \lambda(x+y) \ge \max(\lambda(x),\lambda(y))+1.\end{gathered} \tag{7}$$

A primitive factor pattern $(l_1,\dots,l_t)$ has level in

$$\left[\max\left(\max_i l_i + t - 1,\ \sum_i l_i - t + 1\right),\ \sum_i l_i + t - 1\right]. \tag{8}$$

A representation of a target through level Q has $t \le Q$ and $l_i \le Q-t+1$. If B is empty then D is empty. If $n = q$, then $T = D = \varnothing$ and $W \ge 0$.

Proof. Write $x+\rho = im + r_x$ and $y+\rho = jm + r_y$ with $0 \le r_x, r_y < m$. The extra term in the level of $x+y$ is $\lfloor (r_x + r_y - \rho)/m \rfloor \in \{-1, 0, 1\}$; the strict increments follow from $x, y \ge m$. Iteration gives (8), starting at a factor of largest level for its first lower bound. Positivity of factor levels gives $t \le Q$ and $l_i \le Q - t + 1$. Lemma 1 handles $B = \varnothing$. If $n = q$, all terms of the weighted profile vanish, so $T = D = \varnothing$ and $c = qe - \rho \le ne$. This includes $q = 1$ without dividing by $n - q$. □

Lemma 3 (Eligible pairs)**.** *For a vector $z = (z_1,\dots,z_{Q-1})$ put*

$$P_Q(z) = \sum_{\substack{i<j\\ i+j\le Q+1}} z_i z_j + \sum_{2i\le Q+1} \binom{z_i+1}{2}, \tag{9}$$

$$U_Q(z) = \sum_{\substack{i<j\\ i+j\le Q}} z_i z_j + \sum_{2i\le Q} \binom{z_i+1}{2}. \tag{10}$$

Absent coordinates are zero. With $M = P_q(\alpha)$ and $U = U_q(\alpha)$, $\delta \le M$. Give $x \in T$ weight $w(x) = q - \lambda(x)$; its total is $n - q$. Eligibility of a pair is $w(x) + w(y) \ge q - 1$. If $n - q < q - 1$ (which forces $q \ge 2$, since $n \ge q$), then

$$M \le \left\lfloor \frac{n-q}{\lceil (q-1)/2 \rceil} \right\rfloor; \qquad 2(n-q) < q-1 \implies M = 0.$$

Proof. Each D-value has a pair of proper positive factors in T whose levels sum to at most $q+1$. Choosing one pair per value injects D into the unordered pairs counted by (9), including loops. The weighted identity (5) gives $\sum_{x\in T} w(x) = n - q$. An eligible loop requires weight at least $\lceil (q-1)/2 \rceil$, while distinct vertices have combined weight at most $n-q$. Any pair has weight at most $2(n-q)$. □

Lemma 4 (Missing stars and collisions)**.** *Let E_g consist of the unordered pairs from T whose sum belongs to A. Count a loop once in a vertex's degree. For $u \in R$, $\deg(u) \le K$; if $\deg(u) \ge 2$, then $\delta \le |E_g| - 1$. If u has r eligible neighbors and a actual neighbors, then $|E_g| \le M - r + a$. Consequently an occupied decomposable vertex with at least two eligible neighbors gives $\delta \le M - 1$. For $q \ge 4$,*

$$\delta \le M - \kappa_2 \alpha_{q-1}. \tag{11}$$

Proof. Write $u = x + y$ with positive factors. For any actual neighbor v, the sum $x + v$ is a proper left decomposable subsum of $u + v$; thus $v \mapsto x + v$ injects neighbors into R. Suppose the sum map $E_g \to D$ were injective. Subsum closure gives equal-sum edges $\{v + x, y\}$ and $\{v + y, x\}$. Their equality forces $x = y$ (the other pairing would force $v = 0$). The equal-sum edges $\{v, u\}$ and

$\{x, v+x\}$ must then coincide, forcing $v = x$. Thus there could be only one neighbor, proving the collision claim. Missing eligible pairs give the bound $M - r + a$. If $r \geq 2$, either an eligible edge is missing or $a \geq 2$ gives a collision; either way $\delta \leq M - 1$. These alternatives are not added.

For (11), $u \in R_2$ splits into two A_1 elements. If $v \in A_{q-1}$ and $u + v \in A$, then one of these factors plus v is a proper left subsum at level at least q, impossible. The $\kappa_2 \alpha_{q-1}$ missing pairs are eligible and distinct, and the two layers differ when $q \geq 4$. □

Lemma 5 (One global compression allowance)**.** *Let $\mathcal{C} \subseteq D$ be the set of values having* any *split $z = x + y$ into positive semigroup elements with $\lambda(z) = \lambda(x) + \lambda(y) - 1$. Then $|\mathcal{C}| \leq \rho$. For $z \in D \setminus \mathcal{C}$, every split satisfies $\lambda(z) \geq \lambda(x) + \lambda(y)$. In particular $\delta \leq U + \rho$. If $\delta \leq F$ and $|D \setminus \mathcal{C}| \leq N$, then*

$$q\delta - \rho \leq \min\{qF, (q-1)F + U, (q-1)F + N\}. \tag{12}$$

Proof. For a compressed split of factor levels i, j, their layer lower endpoints and the sum's upper endpoint give

$$(i + j)m \leq z + 2\rho < (i + j)m + \rho.$$

Thus $(z + 2\rho) \bmod m$ belongs to the ρ residues $[0, \rho)$. Distinct Apéry elements have distinct residues, and shifting by 2ρ preserves distinctness. Thus the argument counts the entire set $\mathcal{C}$ once, regardless of the split or its degree. For $z \notin \mathcal{C}$, integrality and (7) give the assertion for every split.

If a decomposable value has no split of level sum at most q, any chosen split has level sum $q + 1$ and is compressed. All other values inject into the U formal pairs. Hence $\delta - \rho \leq U$. Also $\delta - \rho \leq \delta - |\mathcal{C}| = |D \setminus \mathcal{C}| \leq N$. Add either bound to $(q-1)\delta \leq (q-1)F$, and use $\rho \geq 0$ for $q\delta - \rho \leq qF$. □

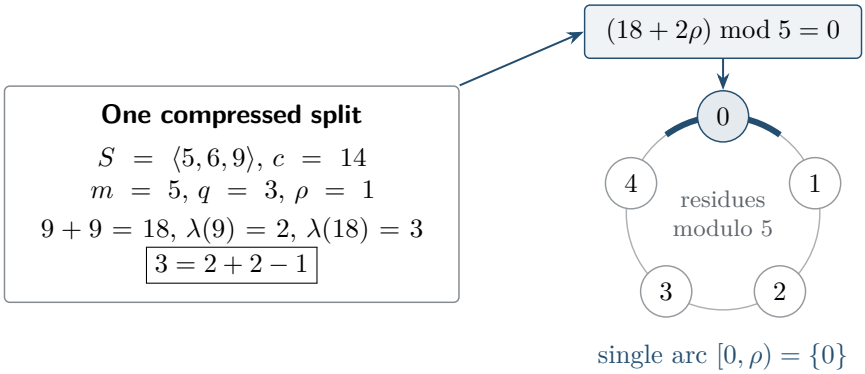


Figure 2. A compressed split lands in the single residue arc $[0, \rho)$ after the common shift $z \mapsto (z + 2\rho) \bmod m$. Distinct Apéry values have distinct residues, so compressed values from every degree and every split share one allowance of ρ slots.

Lemma 6 (Strict endpoints)**.** *Assume $q < n$ and $q\delta - \rho \leq \sigma$. Put*

$$e_{\min} = \max(2, 1 + s), \qquad E = \left\lfloor \frac{\sigma - 1}{n - q} \right\rfloor, \qquad C = qE + \sigma. \tag{13}$$

Any strict violation has $e_{\min} \leq e \leq E$ and $c \leq C$. After applying the $e \leq 3$ input, use $e_{\min} = \max(4, 1 + s)$. Thus $E < e_{\min}$ excludes the branch. With a valid conductor theorem through an inclusive cutoff H, $C \leq H$ also excludes it, and all remaining strict violations have $c \geq H + 1$.

Proof. By (6), $(n-q)e < q\delta - \rho \le \sigma$. These are integers, so $(n-q)e \le \sigma - 1$ and $e \le E$. Also $c = qe + (q\delta - \rho) \le qE + \sigma = C$. Lemma 1 gives $e \ge e_{\min}$; it does not require $e = 1 + s$. The two cutoff conclusions follow, while $q = n$ was settled in Lemma 2. □

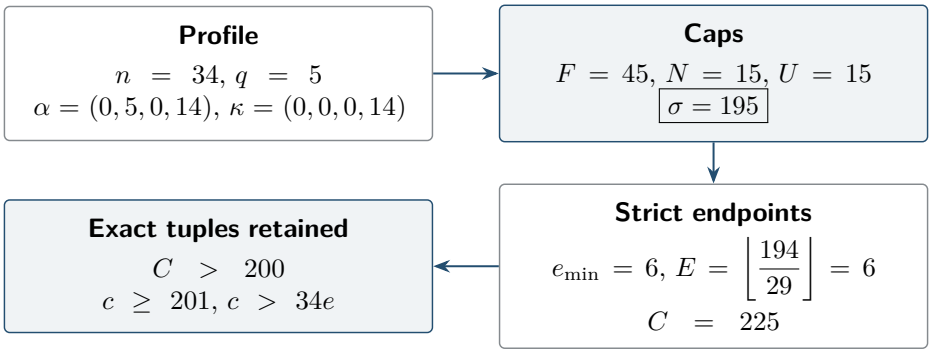


Figure 3. The conductor bound at $n = 34$ (profile event 10563). Here $\sigma = \min\{5F, 4F + U, 4F + N\} = 195$. A strict violation obeys $(n-q)e < q\delta - \rho \le \sigma$, which gives the floor with the essential -1. Here $C > 200$ leaves exact tuples to be examined.

3 A general quadratic conductor reduction

The Apéry identities and compression estimate give a conductor bound depending only on the number of left elements. The ingredients have precedents in Eliahou and Marín-Aragón: the Apéry summand property, the compression allowance, and the weighted left-profile identity appear in [7, Lemma 2.13 and Propositions 2.18 and 2.20], respectively. We give the deduction explicitly. The resulting scalar estimate is a consequence of the profile bounds of Section 2; its relation to those bounds is proved at the end of this section.

Throughout the section, unless stated otherwise, $S \neq \mathbb{N}$. Put

$$t_L = |T| = \sum_{i=1}^{q-1} \alpha_i, \qquad \nu = n - q, \qquad w(x) = q - \lambda(x) \quad (x \in T).$$

Thus t_L counts positive left Apéry elements. By Lemma 1, these weights are positive integers and

$$\sum_{x \in T} w(x) = \nu, \qquad 0 \le t_L \le \nu. \tag{14}$$

Lemma 7 (Weighted pairs and compression)**.** *For every numerical semigroup $S \neq \mathbb{N}$,*

$$q\delta - \rho \le (t_L + 1)\nu. \tag{15}$$

Proof. For each $z \in D$, choose one decomposition $z = x + y$ into positive semigroup elements, recording the unordered pair $\{x, y\}$ and allowing $x = y$. Lemma 1 gives $x, y \in T$. Distinct values of z give distinct selected pairs, since a pair has a unique sum. There are therefore exactly δ selected pairs.

If $i = \lambda(x)$ and $j = \lambda(y)$, Lemma 2 and $\lambda(z) \le q$ give $i + j \le q + 1$. Let b be the number of selected pairs with $i + j = q + 1$. For such a pair, the same level bound forces $\lambda(z) = q = i + j - 1$, so its sum is compressed. These sums are distinct elements of $\mathcal{C}$, whence $b \le |\mathcal{C}| \le \rho$ by Lemma 5.

A boundary pair has weight $w(x) + w(y) = q - 1$; each other selected pair has weight at least q. Consequently,

$$\begin{aligned} q\delta - \rho &\le q\delta - b \\ &\le \sum_{\{x,y\} \text{ selected}} (w(x) + w(y)) \\ &\le \sum_{\substack{\{x,y\} \text{ from } T \\ \text{with repetitions}}} (w(x) + w(y)) = (t_L + 1)\nu. \end{aligned}$$

The third inequality uses positivity of every weight. In the final sum, each vertex occurs in $t_L - 1$ pairs with a different vertex and twice in its loop, so its coefficient is $t_L + 1$; now use (14). If T is empty, all pair sums vanish and the left-hand side is $-\rho \le 0$. The compression allowance counts values, and the selected pairs count each of those values only once. □

Theorem 1 (A structural Wilf bound)**.** *Every numerical semigroup $S \neq \mathbb{N}$ satisfies*

$$W = \nu e - (q\delta - \rho) \ge \nu(e - t_L - 1). \tag{16}$$

In particular, $e \ge t_L + 1$ implies $W \ge 0$. If $W < 0$, then

$$\nu > 0, \qquad 2 \le e \le t_L \le \nu, \qquad c \le nt_L + \nu \le \nu(n + 1). \tag{17}$$

Such a strict violation also satisfies

$$\delta \le \binom{t_L + 1}{2}, \qquad m \le \frac{t_L(t_L + 3)}{2}. \tag{18}$$

Proof. The identity $c = q(e + \delta) - \rho$ from Lemma 1 gives $W = \nu e - (q\delta - \rho)$. Lemma 7 then proves (16). If $\nu = 0$, the same inequality gives $W \ge 0$, also as proved in Lemma 2. Thus $W < 0$ implies $\nu > 0$ and

$$\nu e < q\delta - \rho \le (t_L + 1)\nu.$$

Integrality gives the exact endpoint supplied by this bound:

$$e \le \left\lfloor \frac{(t_L + 1)\nu - 1}{\nu} \right\rfloor = t_L.$$

We have $e \ge 2$ by Lemma 1, and $t_L \le \nu$ by (14). Substituting the embedding-dimension endpoint into the conductor identity yields

$$c = qe + (q\delta - \rho) \le qt_L + (t_L + 1)\nu = nt_L + \nu \le \nu(n + 1).$$

Finally, the selected-pair injection in the preceding proof gives $\delta \le \binom{t_L+1}{2}$. Since $m = e + \delta$ and $e \le t_L$, the claimed multiplicity bound follows. □

Lemma 8 (The shallow and exceptional cases)**.** *The semigroup $\mathbb{N}$ satisfies $W = 0$. Every other numerical semigroup of depth $q \le 2$ satisfies $W \ge 0$.*

Proof. For $S = \mathbb{N}$, the conductor and left cardinality are $c = n = 0$, and $e = 1$, so $W = 0$. A numerical semigroup cannot have conductor 1: in that case it contains both 0 and every positive integer, and its conductor is actually 0. Thus $S \neq \mathbb{N}$ has $c \ge 2$ and $2 \le m \le c$.

If $q = 1$, then $c \le m$, whence $c = m$ and $S = \{0\} \cup [m, \infty)$. Its primitives are exactly $[m, 2m)$: elements of that interval cannot split into two positive elements, and every $x \ge 2m$ equals $m+(x-m)$. Hence $n = 1$, $e = m = c$, and $W = 0$.

Suppose $q = 2$. Every positive left element is primitive, since the sum of two positive elements is at least $2m \ge c$. Thus $\ell_P = n - 1$. Put $L^+ = L \setminus \{0\}$ and

$$d = |(L^+ + L^+) \cap [c, c + m)|.$$

Every decomposition of an element of $[c, c + m)$ uses two left elements: a summand at least c, together with another positive summand, would give a sum at least $c + m$. All m elements of this interval belong to S, so $r_P = m - d$. There are $\binom{n}{2} = n(n-1)/2$ unordered pairs with repetitions from the $n - 1$ elements of L^+. Choosing one pair for each decomposable value in the interval gives $d \le n(n-1)/2$. Therefore

$$W_0 = n\ell_P + 2r_P - c = n(n-1) + 2(m-d) - c \ge 2m - c = \rho \ge 0.$$

Since $W - W_0 = (n - q)r_P \ge 0$, this proves $W \ge 0$. □

Theorem 2 (Quadratic conductor reduction)**.** *Every strict violation of Wilf's inequality satisfies*

$$\begin{gathered} n \ge 5, \qquad 3 \le q \le n-2, \qquad 2 \le e \le t_L \le n - q, \\ 2n \le c \le nt_L + (n-q) \le (n-q)(n+1) \le (n-3)(n+1). \end{gathered} \tag{19}$$

Consequently Wilf's inequality holds for $n \le 4$, and, for each $n \ge 5$, it holds whenever $c > (n-3)(n+1)$.

Proof. Lemma 8 excludes $\mathbb{N}$ and $q \le 2$. Theorem 1 gives $n - q \ge t_L \ge e \ge 2$, so $q \le n - 2$ and $n \ge 5$. The upper conductor bounds follow from (17) and $q \ge 3$.

For the lower bound, the map $x \mapsto c - 1 - x$ sends L injectively into $[0, c) \setminus L$. Indeed, if both x and $c - 1 - x$ belonged to S, their sum $c - 1$ would belong to S, contrary to the definition of the conductor. Thus $n \le c - n$, proving $2n \le c$. The two conclusions follow by contraposition, including $S = \mathbb{N}$ in the first. The endpoint $c = (n-3)(n+1)$ remains in the region of possible strict violations; the guaranteed upper region starts at the next integer. □

Corollary 9 (Finite reduction for bounded left cardinality)**.** *Fix integers $1 \le u \le v$. If $v \le 4$, every numerical semigroup with $u \le n \le v$ satisfies Wilf's inequality. If $v \ge 5$, put*

$$H_v = (v-3)(v+1).$$

The following statements are equivalent.

1. *Every numerical semigroup with $u \le n \le v$ has $W \ge 0$.*
2. *Every numerical semigroup with $u \le n \le v$ and $c \le H_v$ has $W \ge 0$.*
3. *For every integer $2 \le c \le H_v$ and every $L \subseteq [0, c)$ satisfying*

$$\begin{gathered} 0 \in L, \qquad c - 1 \notin L, \qquad u \le |L| \le v, \\ x, y \in L,\ x + y < c \quad \Longrightarrow \quad x + y \in L, \end{gathered} \tag{20}$$

the semigroup $S_{c,L} = L \cup [c, \infty)$ satisfies $|L|\, e(S_{c,L}) \ge c$.

The pairs in the third statement form a finite complete family of canonical representatives of the second statement's semigroups.

Proof. The case $v \leq 4$ is Theorem 2. Assume $v \geq 5$. The first statement implies the second. Conversely, a strict violation in the first statement has $n \geq 5$ and $c \leq (n-3)(n+1)$ by that theorem. The polynomial $h(k) = (k-3)(k+1)$ is increasing on integers $k \geq 5$, since $h(k+1)-h(k) = 2k-1 > 0$. Hence $c \leq H_v$, contradicting the second statement.

For the remaining equivalence, the left part of a numerical semigroup of actual conductor $c \geq 2$ satisfies (20). Conversely those conditions make $S_{c,L}$ an additive submonoid: sums below c are covered by the closure condition, and all sums at least c belong to its tail. It is cofinite, has conductor at most c, and omits $c-1$. The last fact forces its conductor to be exactly c. Its left part is exactly L, so this construction is inverse to taking (c, L). Neither $\mathbb{N}$ nor conductor 1 contributes a semigroup in the stated left-cardinality interval. This proves equivalence with the third statement.

Finally, at fixed c there are at most 2^{c-2} subsets containing 0 and omitting $c-1$, and there are only finitely many values of c. The embedding dimension of each admitted pair also has a finite description. Its multiplicity is $m = \min((L \setminus \{0\}) \cup \{c\}) \leq c$. Every $x \geq c+m$ decomposes as $m + (x-m)$, so all primitives lie in $[1, c+m)$. Membership and all positive summand pairs in this finite interval determine $e(S_{c,L})$ exactly. Thus the third statement concerns a finite family with fully determined invariants. No successful evaluation of this family is assumed. □

For $13 \leq n \leq 37$, the uniform cutoff in the corollary is $H_{37} = 34 \cdot 38 = 1292$. Verification through conductor 200 alone does not discharge this finite reduction: conductors 201 through 1292 are also in its bounded region. At $n = 13$ the scalar cutoff is $H_{13} = 10 \cdot 14 = 140$. The sharper bound $c \leq 84$ obtained from the additional small-embedding-dimension, small-depth and graph inputs in the later application remains stronger. These numerical specializations describe bounds, without asserting any new computation.

Proposition 10 (Comparison with the profile bound)**.** *For $M = P_q(\alpha)$ and $U = U_q(\alpha)$ as in Lemma 3, put*

$$\sigma_{\mathrm{prof}} = \min\{qM, (q-1)M + U\}.$$

Then

$$q\delta - \rho \leq \sigma_{\mathrm{prof}} \leq (t_L + 1)\nu. \tag{21}$$

For $\nu > 0$, the embedding-dimension and conductor endpoints supplied by σ_{prof} in Lemma 6 are at least as strong as those supplied by the scalar bound.

Proof. The selected-pair injection gives $\delta \leq M$. A selected pair whose levels sum to $q+1$ has a compressed sum, and there are at most ρ such sums. All other selected pairs are among the U pairs whose levels sum to at most q. Hence $\delta - \rho \leq U$. Together with $\rho \geq 0$ and $q - 1 \geq 0$, these inequalities give

$$q\delta - \rho \leq qM, \qquad q\delta - \rho = (q-1)\delta + (\delta - \rho) \leq (q-1)M + U.$$

To compare the bounds, view $(q-1)M + U$ as a sum over all eligible pairs from T. A pair whose levels sum to at most q receives coefficient q; a boundary pair whose levels sum to $q+1$ receives coefficient $q-1$. Each coefficient is at most the actual pair weight $w(x) + w(y) = 2q - \lambda(x) - \lambda(y)$. Thus

$$\sigma_{\mathrm{prof}} \leq (q-1)M + U \leq \sum_{\{x,y\} \text{ eligible}} (w(x) + w(y)) \leq (t_L + 1)\nu,$$

where the last inequality extends to all unordered pairs with repetitions and uses the same counting identity as in Lemma 7.

For fixed q and $\nu > 0$, both $\lfloor(\sigma-1)/\nu\rfloor$ and $q\lfloor(\sigma-1)/\nu\rfloor+\sigma$ are nondecreasing functions of σ. Substituting (21) into these two expressions proves the endpoint comparison. □

The quadratic cutoff is therefore a closed-form consequence of the profile and compression estimates. Retaining the profile, and then the shadow and matching constraints developed below, can give smaller finite regions. The finite equivalence above needs no prescribed traversal, certificate format or complexity claim.

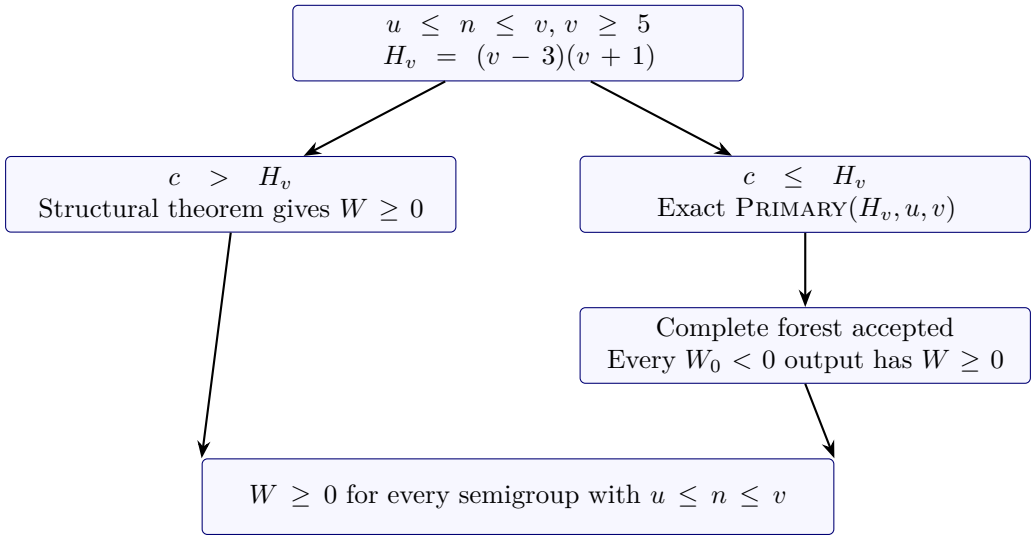


Figure 4. Combining the structural conductor reduction with finite verification. The right-hand conclusion requires an accepted complete forest and successful W tests on every required output. The picture describes the proved implication, not a computation already performed. For $v \le 4$ the elementary structural theorem suffices.

4 Exact enumeration at fixed conductor

Fix integers $C \ge 2$ and $1 \le u \le v$, and write

$$\mathcal{F}(C,u,v) = \{S : c(S) \le C,\ u \le n(S) \le v\}.$$

We give a finite algorithm whose output consists exactly of the members of this family with $W_0 < 0$. The restriction to this output is sufficient for Wilf verification because $W \ge W_0$. The product and large-generator pruning principles come from Bacher [1, §2.2, Lemmas 2.1 and 2.3]. We prove their precise forms for an ordered traversal with fixed *actual* conductor. This traversal retains complete membership below that conductor and includes large-generator extensions in the same tree. Its proof does not establish correctness of Bacher's moving-cutoff algorithms, their shifted fields, or their recorded execution.

4.1 Canonical states and exact reconstruction

The canonical identity of a semigroup with conductor $c \ge 2$ is the pair (c, L), where $L = S \cap [0, c)$. As proved in Corollary 9, a pair represents such a semigroup exactly when $0 \in L$, $c-1 \notin L$, and L

is closed under sums that remain below c. Its semigroup is $L \cup [c, \infty)$. At a traversal node we write $k = |L|$; thus k is the left cardinality of that node and need not yet lie in $[u, v]$.

Lemma 11 (Finite primitive reconstruction)**.** *Let (c, L) be canonical, and put $m = \min((L \setminus \{0\}) \cup \{c\})$. Every primitive is less than $c+m$, and every positive left element is generated by the primitives below c. If $L^+ = L \setminus \{0\}$, then*

$$P \cap [0, c) = L^+ \setminus (L^+ + L^+), \tag{22}$$

$$r_P = m - |(L + L) \cap [c, c + m)|. \tag{23}$$

In particular $e = \ell_P + r_P$ counts all primitives. Moreover

$$q \le k, \qquad W_0 = k\ell_P + qr_P - c, \qquad W - W_0 = (k - q)r_P \ge 0. \tag{24}$$

Proof. If $x \ge c + m$, then $x = m + (x - m)$ is a sum of two positive semigroup elements. For $0 < x < c$, induction on x expresses it in left primitives: if x is decomposable, both positive summands are smaller than x and below c. This also proves (22).

Every decomposition of $x \in [c, c + m)$ uses two positive left elements, since a summand at least c would force the sum to be at least $c + m$. Conversely, every positive pair from L is a decomposition. A zero summand cannot produce such an x from $L + L$, because $x \notin L$. Thus exactly $|(L + L) \cap [c, c + m)|$ of the m threshold elements are decomposable, proving (23). Zero summands must, however, be excluded when testing primitives below c. Finally $0, m, \dots, (q - 1)m$ are q distinct members of L. The displayed formulas for W_0 and $W - W_0$ now follow from the definitions. □

For a finite increasing list G of positive integers below c, set

$$S_c(G) = \langle G \rangle \cup [c, \infty), \qquad L(G, c) = \langle G \rangle \cap [0, c).$$

The empty list is permitted and gives the ordinary semigroup $S_c(\varnothing) = \{0\} \cup [c, \infty)$. Its parameters are

$$m = c, \quad k = q = 1, \quad \ell_P = 0, \quad r_P = e = c, \quad W_0 = W = 0. \tag{25}$$

We account for this semigroup directly at each conductor. The semigroup $\mathbb{N}$ has $c = n = 0$ and $W = 0$; it is outside $\mathcal{F}(C, u, v)$. Conductor 1 cannot occur, as observed in Lemma 8.

Lemma 12 (Ordered primitive prefixes)**.** *Every nonordinary semigroup of conductor c has a unique increasing list $G = (m = g_0 < g_1 < \cdots < g_{\ell_P - 1} < c)$ of left primitives, with $2 \le m < c$, and equals $S_c(G)$. Every nonempty prefix of G defines a semigroup with the same actual conductor c and multiplicity m.*

Proof. Multiplicity 1 would give $\mathbb{N}$. If $m = c$, there are no positive left elements and the semigroup is ordinary. Otherwise m is a left primitive, and Lemma 11 shows that the unique list of all left primitives generates L. A prefix generates a subset of S and has the same full tail. It therefore still omits $c - 1$, so its actual conductor is c. Its least positive element remains m. □

An admitted nonordinary state satisfies the following invariant:

$$\begin{gathered} 2 \le m < c \le C, \qquad G = (m = g_0 < \cdots < g_{\ell_P - 1} < c), \qquad b = \max G, \\ L = L(G, c), \qquad c - 1 \notin L, \qquad G \text{ is exactly the list of left primitives,} \\ k = |L|, \quad q = \lceil c/m \rceil, \quad \rho = qm - c, \quad r_P, e, W_0, W \text{ have their exact defined values.} \end{gathered} \tag{26}$$

Both c and m remain fixed on a tree. Each multiplicity $2 \le m < c$ supplies the seed $G = (m)$. If its closure contains $c-1$, the seed is rejected. Otherwise it satisfies the invariant, with

$$k = q, \qquad \ell_P = 1, \qquad r_P = m-1, \qquad W_0 = qm - c = \rho \ge 0.$$

Indeed its left elements are the multiples of m, and exactly one multiple of m lies in $[c, c+m)$; this multiple is decomposable since $c > m$. These nonnegative seed values do not justify rejecting their descendants.

Lemma 13 (Ordered extension)**.** *Suppose* (26) *holds and $b < a < c$ with $a \notin L$. Append a to G and reconstruct $L' = L(G \parallel a, c)$. The primitives below the fixed cutoff c are exactly the entries of $G \parallel a$, and $k' \ge k+1$. If $c-1 \notin L'$, the child satisfies* (26)*, with $\ell'_P = \ell_P + 1$. If $c-1 \in L'$, no further extension can have actual conductor c.*

Proof. Any new decomposition of an old generator would have to use a in its generator expansion. Its value would then be at least a, larger than that old generator. A decomposition of a cannot use a itself, so would express a using G, contrary to $a \notin L$. Every other positive element of L' has a generator expression; if it is not a listed generator, that expression has at least two positive terms. This proves the primitive assertion. The new member a gives $|L'| \ge |L| + 1$, and $a > m$ preserves multiplicity. Closure and the test at $c-1$ give the canonicality assertion. Subsequent extensions only enlarge L', so cannot remove $c-1$. □

All state reconstruction is finite and exact. One possible definition sets $M[0] = 1$ and, in increasing order of x, uses

$$M[x] = \bigvee_{\substack{g \in G \\ g \le x}} M[x-g] \quad (1 \le x < c), \tag{27}$$

where an empty disjunction is false. Induction on x proves that $M[x] = 1$ exactly when $x \in \langle G \rangle$: remove one generator from a positive expression for the reverse implication. Membership at $x \ge c$ is true. After the canonicality test, the positive-summand test on $1 \le x < c + m$ recovers every primitive and hence all counts. Alternatively (22)–(23) give the same values from pair sums. No condition $\gcd(G) = 1$ is needed, since the appended tail already ensures cofiniteness.

4.2 The two descendant bounds

Lemma 14 (Product bound)**.** *At a state satisfying* (26)*, if*

$$(k+1)(\ell_P + 1) \ge c, \tag{28}$$

every proper canonical descendant in the ordered tree has $W_0 \ge 0$.

Proof. A proper descendant has at least $k+1$ left elements and at least $\ell_P + 1$ left primitives by Lemma 13. Its conductor is still c, and its term qr_P is nonnegative. Thus its Eliahou number is at least $(k+1)(\ell_P+1) - c \ge 0$. The statement concerns proper descendants and makes no assertion about the current node's sign. □

Lemma 15 (Large-generator suffix)**.** *At an admitted node with $\ell_P \ge 2$, put*

$$\gamma_1 = g_1, \qquad h = c + m - \gamma_1, \qquad \Delta = k + \ell_P + 1 - q. \tag{29}$$

If a canonical child appends $a \ge h$, then

$$r'_P \ge r_P - 1, \qquad W'_0 - W_0 \ge \Delta \ge \ell_P + 1 > 0. \tag{30}$$

If $W_0 + \Delta \geq 0$, every canonical descendant whose next appended generator is at least h has nonnegative Eliahou number. Both assertions include equality $a = h$.

Proof. An old decomposable threshold element remains decomposable. If $x \in [c, c+m)$ becomes newly decomposable, expand its two positive left summands in $G \parallel a$. This expression must use a. It cannot also use an old generator at least γ_1, since $a + \gamma_1 \geq c + m$. It cannot use a twice, since $a > \gamma_1$ and hence $2a > a + \gamma_1 \geq c + m$. The expression therefore has the form $x = a + jm$ for some $j \geq 1$. A half-open interval of length m contains at most one integer in this residue class, so at most one right primitive is lost. At $a = h$, the excluded sum $a + \gamma_1 = c + m$ is still outside the interval.

Using $\ell'_P = \ell_P + 1$, $k' \geq k + 1$, and the fact that q does not change, we obtain

$$W'_0 - W_0 = (k' - k)(\ell_P + 1) + k + q(r'_P - r_P) \geq k + \ell_P + 1 - q = \Delta.$$

The inequality $k \geq q$ proves its strict positivity. Every later generator is larger than a, whereas c, m, γ_1 stay fixed. The same argument consequently applies at each later canonical step. When $W_0 + \Delta \geq 0$, the first child is nonnegative and every further step increases its Eliahou number. This proves the suffix assertion for any finite number of extensions. If a prefix is noncanonical, inclusion precludes a canonical descendant altogether. □

There is also the elementary upper-cardinality rule: a state with $k > v$ has no descendant in $\mathcal{F}(C, u, v)$, because left sets only grow. There is no lower-cardinality rule; a prefix with $k < u$ may reach the target interval. Nor is $W_0 \geq 0$ at the current node by itself a reason to discard descendants.

4.3 Traversal and exactness theorem

The following pseudocode fixes the rule order. Its revision is `fixed-c-ordered-left-dfs-v1`; u, v are the lower and upper output bounds. The operation `reconstruct` first computes L and the canonicality test, and computes the remaining fields when the state is canonical. Every state is an immutable mathematical value: a child call leaves the parent's fields and pending candidates unchanged. In the pseudocode, `ell`, `r`, and `W0` denote ℓ_P, r_P, W_0.

```
PRIMARY(C,u,v):
  require C >= 2 and 1 <= u <= v
  for c = 2,...,C:
    account for G = empty, m = c, W0 = W = 0
    for m = 2,...,c-1:
      s := reconstruct(c,m,(m))
      if c-1 in s.L: record NONCANONICAL_SEED
      else: VISIT(s)

VISIT(s):
  if k > v: record RANK_UPPER; return
  if u <= k <= v and W0 < 0: emit the exact current record
  if (k+1)*(ell+1) >= c: record PRODUCT; return
  for a = max(G)+1,...,c-1, in increasing order:
    if a in L: account for ALREADY_GENERATED; continue
    if ell >= 2:
      h := c+m-G[1]; Delta := k+ell+1-q
      if a >= h and W0+Delta >= 0:
        record LARGE_TAIL starting at a; return
    t := reconstruct(c,m,G appended with a)
```

```
    if c-1 in t.L: record NONCANONICAL_CHILD; continue
    VISIT(t)
  record CANDIDATES_EXHAUSTED
```

Here $G[1]$ is the second entry, with indices starting at zero. The exact current output has identity (c, L) and fields

$$(c, L, G, m, k, \ell_P, r_P, e, q, \rho, W_0, W). \tag{31}$$

The value of W is reported and is not an output filter. Both descendant bounds follow the current output step; the upper-cardinality exit precedes it only because $k > v$ makes output impossible. In a one-generator seed the second-generator gate is undefined and is never evaluated. Empty multiplicity or candidate ranges simply contribute no iterations. No additional gate involving $c \le 3m$, $c + m \le 3\gamma_1$, a bound on ℓ_P, small left cardinality, or any conjecture is used.

Theorem 3 (Exact primary enumeration)**.** *For every $C \ge 2$ and $1 \le u \le v$, `PRIMARY(C,u,v)` terminates and emits exactly the canonical records of the semigroups in $\mathcal{F}(C, u, v)$ with $W_0 < 0$, each once. Every field in (31), including the full embedding dimension, is exact.*

Proof. Seed reconstruction and Lemma 13 establish (26) at each call to `VISIT`. Hence every output is canonical, has actual conductor at most C, satisfies the output bounds, and has the stated counts by Lemma 11. This proves soundness.

For completeness, fix $S \in \mathcal{F}(C, u, v)$ with $W_0(S) < 0$. Equation (25) excludes the ordinary case. Its unique increasing primitive list has the canonical prefix path of Lemma 12. Every prefix left set is contained in the target left set, so no prefix contains $c - 1$ or has cardinality greater than v. Each next target primitive is greater than the preceding maximum and absent from the prefix closure; otherwise it would be generated by smaller target elements. Thus neither an already-generated candidate nor a noncanonical rejection removes this path. At a proper prefix, a product exit would force $W_0(S) \ge 0$ by Lemma 14. If a large-tail exit occurred before its next target generator, that generator and every later target generator would be at least h; Lemma 15 would again force $W_0(S) \ge 0$. These contradictions show that the path reaches the whole list. The output step then records S before any descendant exit.

For uniqueness, the actual conductor and multiplicity are uniquely determined by S, as is its full left primitive list. An ordered list has only one path from its seed, obtained by appending its successive entries. Each candidate is used once at its parent, so no canonical identity can be emitted twice.

Finally, every child strictly decreases the nonnegative integer $c - 1 - \max G$. Reconstruction and all candidate loops are finite, so induction on this integer proves termination at a node. There are finitely many seeds. More explicitly, at fixed (c, m) every constructed label is (m) followed by a subset of $\{m + 1, \dots, c - 1\}$. Including rejected children and the ordinary accounting, the number of possible labels at conductor c is at most

$$1 + \sum_{m=2}^{c-1} 2^{c-m-1} = 2^{c-2}.$$

Summing gives at most $2^{C-1} - 1$ labels. This is a finite upper bound; it is not a measured traversal size. □

A second finite description enumerates all subsets of $[0, c)$ containing 0 and omitting $c - 1$, tests the canonical closure condition, and applies the same output predicate. Canonical uniqueness and Theorem 3 prove equality of its exact (c, L) output with that of the primary algorithm. This

mathematical comparison specifies a reference for later validation, without asserting that either enumeration has been executed.

4.4 An exact bit representation

Finite sets may equivalently be represented by nonnegative integers $B_X = \sum_{x \in X} 2^x$, where bit x denotes the actual integer x. In particular bit zero belongs to B_L, and its population count is k. A logical left shift and bitwise union give

$$B_{L+L} = \bigvee_{x \in L} (B_L \ll x), \qquad B_{L^+ + L^+} = \bigvee_{x \in L^+} (B_{L^+} \ll x). \tag{32}$$

Indeed bit z occurs exactly when $z = x + y$ for an allowed pair. Union is bitwise OR, rather than integer addition, so multiplicities of representations cause no carries. The membership field has c bits and each sumset field has $2c - 1$ bits, indexed from 0 through $2c - 2$; queries beyond this range are zero.

The exact closure after adjoining a has the representation

$$B_{L'} = \left(\bigvee_{j=0}^{\lfloor (c-1)/a \rfloor} (B_L \ll ja) \right) \,\&\, (2^c - 1). \tag{33}$$

Its support is $(L + \mathbb{N}a) \cap [0, c)$, which equals $\langle G \| a \rangle \cap [0, c)$. An expression below c has its old-generator part below c and its coefficient of a at most $\lfloor (c-1)/a \rfloor$, proving both inclusions. Clipping at c discards only values in the already present tail; nonnegative addition can never bring these values back below c. The next sumset is reconstructed from the entire retained L'. No retained lower bits are discarded, and shifts never wrap around.

The quantities $q = \lfloor (c + m - 1)/m \rfloor$ and $\rho = qm - c$ use exact integer arithmetic; q is constant within a tree. The numbers W_0, W are signed. These identities establish correspondence for full finite sets and arbitrary-precision bit fields. Any fixed-width, shifted, truncated, vectorized, or mutable implementation must separately justify its integer ranges and its preservation of these state semantics.

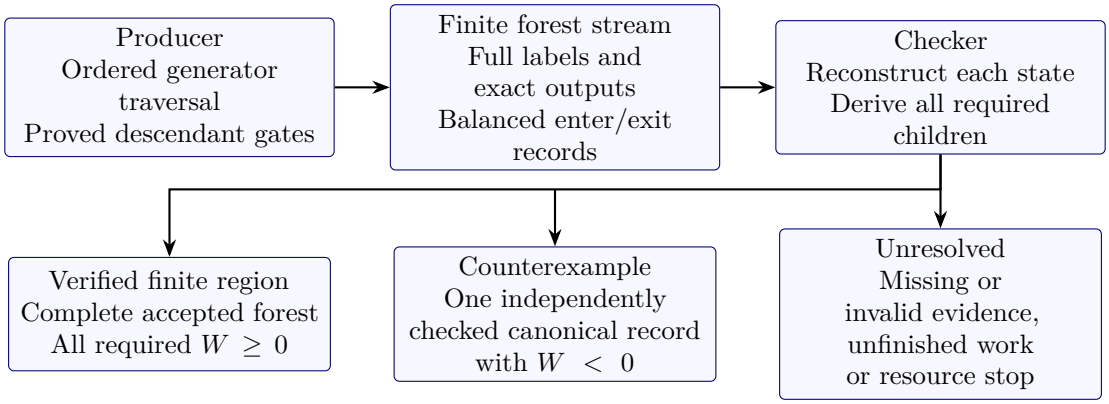


Figure 5. The producer supplies evidence; the checker derives coverage from the required roots and children, rather than trusting a producer count. A single valid counterexample does not require a completed forest. Malformed or incomplete evidence gives no mathematical negative conclusion. These are specification-level roles; separate implementations have not been run.

5 Certificates of complete finite verification

A list of valid outputs does not establish that every required output has been found. We therefore specify a finite forest whose local checks establish coverage of the primary traversal. The checker reconstructs the mathematical states from their labels; supplied counts or completion assertions are not premises.

5.1 Header, roots and node normal form

The certificate header binds the integers $C \geq 2$, $1 \leq u \leq v$, and the state and integer representations. Its algorithm revision is

$$\texttt{fixed-c-ordered-left-dfs-v1}.$$

Its exact output predicate is

$$u \leq k \leq v \quad \text{and} \quad W_0 < 0. \tag{34}$$

It binds the gate formulas and their order in Section 4, together with the exceptional-case conventions (25) and $W(\mathbb{N}) = 0$. In a concrete bundle the producer and checker source revisions and the retained specification must also be identified. These bindings identify which statement is checked; a digest alone proves no mathematical claim.

There is exactly one nonordinary tree for each pair in

$$\mathcal{R}_C = \{(c, m) : 2 \leq c \leq C,\ 2 \leq m < c\}, \tag{35}$$

with root label $(c, m, (m))$. The ordinary semigroup at each c is accounted for by (25), without a separate tree. Thus for $C = 2$ the nonordinary forest is empty, but its conductor accounting is still complete. A node label is (c, m, G), where G is a nonempty increasing list beginning at m with entries below c. Every edge appends exactly one integer to its parent's list.

The checker independently reconstructs $L(G, c)$ by (27). It checks the successive-gap property: each entry of G after the first is absent from the closure of its predecessors. Lemma 13 then proves that, whenever $c - 1 \notin L$, the list is exactly the left primitive list. For canonical states, the checker uses membership in $L \cup [c, \infty)$ and declares $1 \leq x < c + m$ primitive precisely when

$$x \in S \quad \text{and} \quad \text{there is no } y \in \{1, \ldots, x-1\} \text{ with } y, x - y \in S. \tag{36}$$

It counts both left and right primitives, checks that the former count is $|G|$, and derives e, q, ρ, W_0, W. This test agrees with the producer's pair-sum formulas by Lemma 11; it requires no supplied primitive counts. Separate mathematical descriptions do not by themselves establish independence of future implementations.

The following ordered normal form is mandatory at each node.

1. If $c - 1 \in L$, the node has reason `NONCANONICAL`, no output and no children. The remaining cases assume $c - 1 \notin L$.

2. If $k > v$, the node has reason `RANK_UPPER`, no output and no children. The remaining cases assume $k \leq v$.

3. An output is present if and only if (34) holds. Its canonical identity and all fields in (31) must agree with reconstruction. Equality of aggregate counts is insufficient.

4. If $(k+1)(\ell_P+1) \geq c$, the node has reason `PRODUCT` and no children. Its output, if required in the previous step, remains present.

5. Otherwise derive the ascending list of gaps

$$\mathcal{A}_G = \{a : \max G < a < c,\ a \notin L\}. \tag{37}$$

If $\ell_P \geq 2$, $W_0 + \Delta \geq 0$, and $\mathcal{A}_G$ has a member at least h, let a_0 be its least such member, using h, Δ from (29). The declared tail must then be exactly

$$\mathcal{T}_G = \{a \in \mathcal{A}_G : a \geq a_0\}, \qquad \texttt{TAIL} = (\gamma_1, h, \Delta, a_0). \tag{38}$$

In all other cases $\mathcal{T}_G$ is empty and no tail record is allowed. Require exactly one child $(c, m, G \,\|\, a)$ for every $a \in \mathcal{A}_G \setminus \mathcal{T}_G$, in increasing order, and no other children. If there are no children and no tail, the node has reason `EXHAUSTED`. Otherwise it has an expansion record, including the exact tail data when the tail is nonempty.

This includes rejected gap children as explicit terminal nodes. The reason `NONCANONICAL_SEED` becomes `NONCANONICAL`. The same alias is used for `NONCANONICAL_CHILD`. The reason `LARGE_TAIL` becomes the derived `TAIL`. When no children or tail remain, `CANDIDATES_EXHAUSTED` becomes `EXHAUSTED`. A node whose candidate loop finishes after visiting children is an expansion with those children and no tail. There is no child for `ALREADY_GENERATED`, because (37) excludes generated values. In particular the tail begins at the first eligible *gap*, even if intervening integer candidates were already generated. Equality $a_0 = h$ is allowed by Lemma 15.

5.2 An explicit forest encoding

The logical object is a finite forest, with no shared subtrees. One baseline encoding is a sequence of balanced `ENTER` and `EXIT` records. An `ENTER` carries the full label (c, m, G), its terminal or expansion descriptor, the tail tuple if required, and the output record if required. It is followed by the encodings of its children in the order above and then one `EXIT`. Roots are ordered lexicographically by (c, m). The header states this grammar and requires the forest to end after the last required root.

All lists are length-prefixed and strictly increasing where specified. In the baseline output payload, L is its sorted list of elements, including zero, and G is its full sorted primitive list; the remaining fields are the exact integers in (31). For definiteness, put $d = \lfloor \log_2(z+1) \rfloor + 1$ for a nonnegative integer z. Its code is 1^d0 followed by the d-bit binary expansion of $z+1$. Signed integers have a sign bit followed by this code for their magnitude; the sign bit is zero for nonnegative integers, so zero has only one permitted encoding. Record tags belong to a fixed finite alphabet with a fixed prefix code. These conventions give an unambiguous finite bit string. The full c-bit mask of L is an alternative payload, with its different size accounted for separately; it is not implicit in the sorted-list baseline.

On entry the parser compares the label with the next required root or the next required child of its current stack frame. It reconstructs the state and determines its ordered expected children from the normal form. The frame retains that state and a cursor for the next required child. An `EXIT` is accepted only when all those children have closed. An entry when no child is expected, a premature exit, an unmatched exit, or an unfinished stack is rejected. No record after the last root is accepted. These rules check missing or repeated roots, missing or duplicate children, malformed edges and disconnected data. The strict extension of G and the nested grammar exclude cycles; there is no pointer or DAG interpretation. Retaining the parent's reconstructed frame preserves every pending

sibling across child calls. This is a precise encoding and parsing specification, not a claim about an executed streaming checker.

Root sharding has the same semantics. A manifest must partition $\mathcal{R}_C$ exactly once, with every required pair assigned to exactly one shard. Each shard contains the closed trees for precisely its assigned roots, in the stated order, and uses the same header parameters and revision. Global acceptance requires acceptance of every shard and the exact partition check. Formula accounting for ordinary semigroups is shared by the global header. A pending shard leaves the global result incomplete, regardless of the counts or digests of completed shards.

Theorem 4 (Exhaustive-forest checker)**.** *For every admissible C, u, v, an accepted finite certificate in this normal form has exactly the output of* `PRIMARY(C,u,v)`*, with each canonical identity occurring once and all invariant values correct. If every output also passes the exact test $W \geq 0$, then*

$$\text{every } S \in \mathcal{F}(C, u, v) \text{ satisfies } W(S) \geq 0. \tag{39}$$

Conversely, the completed finite abstract traversal has a certificate in this normal form; the additional W tests need not all succeed.

Proof. For a recorded output, closure reconstruction and the missing value $c - 1$ give its canonical semigroup. The primitive test (36) recovers all primitives by Lemma 11, so every reported invariant and the output predicate are exact. The increasing primitive list and its unique root and parent paths give uniqueness, as in Theorem 3.

To prove completeness without relying on a producer's assertion, fix a target $S \in \mathcal{F}(C, u, v)$ with $W_0(S) < 0$. Its ordered primitive path begins at one of the required roots. Its canonical prefixes cannot terminate as `NONCANONICAL`, and their left cardinalities do not exceed $n(S) \leq v$, so none terminates as `RANK_UPPER`. At a proper prefix, `PRODUCT` would contradict Lemma 14. If its next generator belonged to the declared tail, Lemma 15 would similarly contradict $W_0(S) < 0$, including when that generator equals h. In every other case the next generator is an eligible gap outside the tail and is therefore a required child. The parser's closure rules ensure that this child is present. Induction through the finite primitive list reaches the target itself, where step 3 requires its exact output. Thus every required record is present. Soundness and Theorem 3 prove equality with the primary output.

For (39), a target with $W_0 \geq 0$ has $W \geq 0$ by (24). Every other target is in the complete output and passes $W \geq 0$ by the additional hypothesis. Finally, termination of the abstract traversal allows one to retain all its roots, constructed gap children, exact outputs and derived exits. Normalize them as above and encode each finite tree by balanced entry and exit. This gives an accepted forest; its separately computed W values determine whether the additional tests succeed. No assumption on those signs was used to construct the certificate. □

We use `VERIFIED(C,u,v)` only for a complete accepted forest whose required outputs all have $W \geq 0$. One independently reconstructed canonical record with the target bounds and $W < 0$ is a counterexample, even if the forest is incomplete. Other outcomes, including malformed records, missing children, unfinished shards or a resource stop, leave the universal verification unresolved. Failure to parse a proposed certificate is not a counterexample to Wilf's inequality. No certificate or successful new computation is asserted in this section.

5.3 Universal conductor and bounded-cardinality consequences

Corollary 16 (All left cardinalities at bounded conductor)**.** *For every $C \geq 2$, the condition*

$$\texttt{VERIFIED}(C, 1, \lfloor C/2 \rfloor)$$

implies Wilf's inequality for every numerical semigroup of conductor at most C.

Proof. The semigroup $\mathbb{N}$ has $W = 0$. For any other semigroup, reflection $x \mapsto c - 1 - x$ injects L into $[0, c) \setminus L$: otherwise two elements of S would sum to its gap $c - 1$. Consequently $1 \le n \le \lfloor c/2 \rfloor \le \lfloor C/2 \rfloor$, and Theorem 4 applies. □

In particular, `VERIFIED(200,13,37)` would establish exactly the conductor-200 result restricted to $13 \le n \le 37$. The all-cardinality conclusion instead follows from `VERIFIED(200,1,100)`. These are different certificate scopes; the first does not establish the second. Neither antecedent is claimed here.

Corollary 17 (Certificate form of the finite reduction)**.** *Let $1 \le u \le v$. If $v \le 4$, Wilf's inequality holds throughout this interval. If $v \ge 5$ and $H_v = (v - 3)(v + 1)$, the following statements are equivalent:*

1. *Every numerical semigroup with $u \le n \le v$ satisfies Wilf's inequality.*
2. *There exists a complete accepted forest for `PRIMARY`(H_v, u, v) whose required outputs all have $W \ge 0$.*

Proof. The small-cardinality assertion is Theorem 2. For $v \ge 5$, the second statement and Theorem 4 prove Wilf's inequality in conductor at most H_v; Corollary 9 then proves it for the whole interval. Conversely, the first statement implies that every primary output at these parameters has $W \ge 0$. The finite abstract traversal has an accepted forest by Theorem 4, proving the second statement. □

This equivalence concerns existence of a finite certificate with the specified successful checks. It supplies a method for the remaining finite domain and does not assume a successful run. The retained 13–37 application later in the paper continues to use its stated historical computational inputs.

6 Representation and the cost of finite verification

The inequality $ne \ge c$ is inexpensive to test when its three invariants are already known. Obtaining those invariants, listing semigroups, and certifying that a family has been exhausted are different computational tasks. We give explicit upper bounds for each. The bounds concern the specified mathematical procedures; they are not measurements of the recorded profile computation in Section 11.

6.1 Input and output conventions

All integer inputs are binary. We use a deterministic random-access bit-cost model: reading, writing, adding or comparing a b-bit integer, including its address, costs $O(b)$; schoolbook multiplication and division cost $O(b^2)$. A packed string of h bits can be scanned or combined by bitwise OR in $O(h)$ time. An addressed bit operation is charged $O(\log(h + 2))$. Space is measured in bits. Retained input and output are included when stated; streamed output still contributes its length to time. These conventions specify the model, rather than a unit-cost machine with unbounded integer words.

We distinguish three inputs: validated invariants (n, e, c); a claimed canonical pair (c, L); and an arbitrary positive generator list. For a family, the input is (C, u, v) and the output may include both the W_0-negative records and a complete traversal certificate. An arbitrary implementation journal is not this certificate and has whatever additional length its format requires.

Proposition 18 (Arithmetic and sparse extraction)**.** *Given valid b-bit invariants n, e, c, the value and sign of $W = ne - c$ can be computed in $O(b^2)$ time and $O(b)$ additional space. Suppose instead that $c \geq 2$ and L is a list of k integers claimed to be the left part at conductor c. Put $\lambda_{\mathrm{bit}} = \lceil \log_2(c+2) \rceil$ and let I be the input length. Validation and extraction of n, m, e, q, ρ, W_0, W, with a compact exact primitive description, take*

$$O(I + k^2 \log(k+1)\lambda_{\mathrm{bit}} + \lambda_{\mathrm{bit}}^2) \quad \textit{time}, \qquad O((k+1)^2 \lambda_{\mathrm{bit}}) \quad \textit{additional bits}.$$

An explicit list of all primitives adds $O(e\lambda_{\mathrm{bit}})$ output time and bits.

Proof. The first assertion follows by multiplication and subtraction and does not validate the supplied invariants. For the second, reject $k = 0$ or $k > c$ before allocating pair data. Sort the list; reject repetitions, entries outside $[0, c)$, absence of zero, or presence of $c - 1$. Let

$$L^+ = L \setminus \{0\}, \qquad D_+ = L^+ + L^+.$$

Generate, sort and deduplicate the at most $k(k-1)/2$ unordered positive pair sums, with repetitions allowed. Check that every such sum below c belongs to L. By (20), these are exactly the canonicality conditions. Set $m = \min L^+$ if L^+ is nonempty, and $m = c$ otherwise. The primitives below c are $L^+ \setminus D_+$. A value in $[c, c+m)$ is decomposable exactly when it belongs to D_+: a positive summand at least c would force a sum at least $c + m$. All primitives are below $c + m$. Hence

$$\ell_P = |L^+ \setminus D_+|, \qquad r_P = m - |D_+ \cap [c, c+m)|, \qquad e = \ell_P + r_P.$$

The exact primitive description is the displayed left set together with $[c, c+m)$ minus its sorted forbidden values. This avoids expanding a long interval merely to count its members. The other invariants follow from their definitions. Pair sums have $O(\lambda_{\mathrm{bit}})$ bits, and sorting and merging give the stated time and space bounds. The scalar arithmetic uses $O(\lambda_{\mathrm{bit}})$-bit integers: $k \leq c$ and $e \leq m \leq c$. Indeed distinct primitives other than m have distinct nonzero residues modulo m, since the larger of two in one residue would decompose. Walking the complement intervals emits the explicit primitives at the claimed output cost. Invalid overlong entries can be rejected while reading their I bits. □

For the ordinary semigroup, $L = \{0\}$ and $m = e = c$. Its sparse description has $O(\log c)$ bits, whereas the explicit primitive list $c, c+1, \ldots, 2c-1$ needs $\Theta(c \log c)$ bits in an ordinary list encoding. Thus polynomial extraction from sparse input does not imply polynomial time for every expanded output. A dense alternative validates a c-bit membership mask and tests decomposability on $[0, c+m)$ by positive pairs in $O(c^2 \lambda_{\mathrm{bit}})$ time and $O(c)$ working bits, before optional output. The case $S = \mathbb{N}$ is handled separately by $c = n = 0, e = 1$.

6.2 Extraction from generators

The Nijenhuis–Dijkstra residue shortest-path method is classical; see [2, §§1–2] for the Frobenius graph and its relation to representability, and [3] for the shortest-path algorithm. We include the argument to make the input promise and output costs explicit.

Let the nonempty input list have d positive integers, minimum m, maximum $a_{\max}$ and total bit length I. Minimality is not assumed. Check positivity and gcd one, and remove repetitions. The original d still bounds the resulting list length. If $m = 1$, the semigroup is $\mathbb{N}$. Otherwise put

$$B_{\mathrm{bit}} = \left\lceil \log_2\big((d+2)(m+2)(a_{\max}+2)\big) \right\rceil.$$

On residues modulo m, give each generator a the directed edges $z \longrightarrow z + a \pmod m$ of weight a. Generate edges when needed; they need not be stored. Let w_z be the distance from zero.

Proposition 19 (Generator-list extraction)**.** *For $m \geq 2$, the Apéry table, all the above invariants and the minimal generator set can be computed in*

$$O(I + dB_{\text{bit}}^3 + (dm \log(m+1) + d \log(d+1))B_{\text{bit}} + (d^2 + m)B_{\text{bit}}^2 + Z_{\text{extra}})$$

time and $O(I + (m + d)B_{\text{bit}})$ resident bits, before additional retained output. Here Z_{extra} is the length of any further requested output.

Proof. Path weights are precisely sums of generators in the endpoint's residue. The nonnegative residue combinations form a subgroup, since in a finite group the inverse of an element is a nonnegative multiple of it. Gcd one therefore makes every residue reachable. It follows that w_z is the least semigroup element in residue z. Positive weights allow deletion of repeated-vertex cycles from a shortest path, so $w_z \leq (m-1)a_{\max}$; a tested relaxation is at most $ma_{\max}$.

Use Dijkstra's algorithm with an indexed binary heap containing at most m tentative labels. When the smallest label is finalized, a shorter path would have a first edge leaving the finalized set; that edge would already give a tentative label no larger than the supposed shorter distance, a contradiction. This proves exactness, including ties. The table gives

$$\begin{gathered} x \in S \iff x \geq w_{x \bmod m} \quad (x \geq 0), \qquad c = \max_z w_z - m + 1, \\ g = \sum_{z=0}^{m-1} \frac{w_z - z}{m}, \qquad n = c - g. \end{gathered} \tag{40}$$

For each residue, its gaps are $z, z+m, \ldots, w_z - m$, counted by $(w_z - z)/m$; the last gap is $\max_z w_z - m$. This proves the formulas.

Every primitive occurs in the original list, since any expression for a primitive has exactly one positive term. A listed a_i is decomposable if and only if some listed $a_j < a_i$ satisfies

$$a_i - a_j \geq w_{(a_i - a_j) \bmod m}.$$

The reverse implication supplies a positive decomposition. For the forward implication, expand one positive summand in a decomposition and remove one generator term; the remaining positive value belongs to S. Testing all ordered pairs finds the full minimal generating set.

Every value and address used above has $O(B_{\text{bit}})$ bits. There are at most dm relaxations and heap updates; each update uses $O(\log(m+1))$ comparisons or moves. Precompute generator residues, so an edge endpoint uses addition and at most one modular subtraction. The d^2 membership tests and m genus divisions use schoolbook arithmetic. Sorting costs $O(d \log(d+1)B_{\text{bit}})$; elementary Euclidean gcd tests have the conservative bound $O(dB_{\text{bit}}^3)$. An implicit graph and indexed heap use the stated space. Reading and writing the specified data account for the remaining terms. □

The bound is pseudopolynomial in the numeric multiplicity m, and is not asserted polynomial in I. Nor is the chosen heap algorithm claimed optimal; [2] studies faster residue methods. An explicit Apéry table already has m entries. In fixed B_{bit}-bit fields it occupies $\Theta(mB_{\text{bit}})$ bits. A dense membership table below c adds $O(cB_{\text{bit}}^2)$ time and c output bits. Alternatively, merging the m residue progressions below c lists L in $O((m+n)\log(m+1)B_{\text{bit}})$ additional time, $O(mB_{\text{bit}})$ working space and $O(nB_{\text{bit}})$ output bits. These representation costs cannot be discarded because the parameters were supplied in binary.

6.3 Traversal and certificate bounds

Consider PRIMARY(C, u, v) and the normal-form checker of Section 5. Let I_2 be the binary header length, and put

$$v_* = \min(v, C), \qquad \lambda_{\mathrm{bit}} = \lceil \log_2(C+2) \rceil, \qquad h = \min(v_*, \max(0, C-2)).$$

Node comparisons may replace v by v_* and u by $\min(u, C+1)$, because $k \le c \le C$. The original header remains part of the input. Let V count every constructed nonordinary node, including noncanonical and upper-cardinality rejections, and let J count emitted $W_0 < 0$ records. These are traversal quantities, not values inferred from an output count alone.

Lemma 20 (Constructed labels)**.** *The number of nonordinary constructed nodes satisfies*

$$V \le U(C, v) := \sum_{j=1}^{\min(v, C-2)} \binom{C-1}{j+1}, \qquad V + C - 1 \le 2^{C-1} - 1. \tag{41}$$

No recursion path has more than h simultaneous nonordinary states.

Proof. At an admitted node, $\ell_P \le k - 1 \le v - 1$, since zero and its distinct positive primitives belong to L. An attempted child can have $\ell_P = v$ and then terminate by rejection. At fixed c every constructed list therefore has between 1 and $\min(v, c-2)$ elements in $\{2, \dots, c-1\}$. Its first element determines its root and its sorted prefixes determine its unique path. Thus

$$V \le \sum_{c=3}^{C} \sum_{j=1}^{\min(v, c-2)} \binom{c-2}{j} = \sum_{j=1}^{\min(v, C-2)} \binom{C-1}{j+1},$$

by the hockey-stick identity. Including the $C-1$ ordinary accounts and summing over all subsets gives the second bound. Each edge increases the list length by one, giving the stack bound. Using $v - 1$ in the bound for *all* nodes would omit rejected children. □

Theorem 5 (One concrete resource realization)**.** *There are exact producer and checker designs with time*

$$O(I_2 + VC^2\lambda_{\mathrm{bit}} + C\lambda_{\mathrm{bit}} + Z) \tag{42}$$

and working space, excluding retained output,

$$O(I_2 + hC + h^2\lambda_{\mathrm{bit}} + \lambda_{\mathrm{bit}}). \tag{43}$$

For full-label enter/exit records and sorted-list left-set outputs, one complete certificate has length

$$Z = O\big(I_2 + (V + C)(h+1)\lambda_{\mathrm{bit}} + J(v_* + 1)\lambda_{\mathrm{bit}}\big). \tag{44}$$

These bounds describe a completed conforming traversal and its evidence.

Proof. Compute closure afresh by the increasing-value Boolean recurrence in Section 4. At list length ℓ_P it uses $O(c\ell_P)$ addressed bit operations. Dense arrays for L, $L + L$ and $L^+ + L^+$ have lengths at most $c, 2c-1, 2c-1$; constructing pair sums takes $O(c^2)$ such operations. Independent positive-pair primitive reconstruction below $c + m$ has the same bound. Candidate scanning costs $O(c)$ per expanded node, and scalar arithmetic $O(\lambda_{\mathrm{bit}}^2)$ per node. Since $\ell_P \le c$ and $\lambda_{\mathrm{bit}} = O(C)$, these costs give (42), including ordinary accounts and serialized evidence. A sharper form sums $O(c_x^2\lambda_{\mathrm{bit}})$ over constructed nodes x.

The checker verifies the root and each extension against the reconstructed parent's gap set, then independently reconstructs the child's full state. It need not recompute every prefix within every node. The parent keeps its arrays, list and next-candidate cursor while visiting a child. One state needs $O(c + \ell_P \lambda_{\text{bit}})$ bits. Summing over at most h ancestors, including a just-constructed rejected child, yields $O(hC + h^2 \lambda_{\text{bit}})$; shared pair-sum and arithmetic scratch fit this bound with the final $O(\lambda_{\text{bit}})$ term.

The full-label stream records the required root order, each node's increasing G, its reason or tail fields, its exact output if required, and balanced exits. Each small field needs $O(\lambda_{\text{bit}})$ bits, including signed W_0, W, whose magnitudes are $O(C^2)$. Each output list has $k \le v_*$ entries and $|G| \le v_*$, so its length is $O((v_* + 1)\lambda_{\text{bit}})$. The original possibly longer bounds u, v occur only in the header. This proves (44). The parser retains reconstructed ancestor frames and the next expected child. It checks the normal form, exact output equality and a completed root inventory; it rejects premature exits, duplicates, extra children and unfinished streams. No backreference or trusted producer count is needed. Thus it uses the same stack bound. Retaining the whole certificate adds Z bits; streaming it does not remove its time cost. □

Dense left-set payloads instead add $O(JC)$ bits. Arbitrarily padded or malformed streams are not subject to the size estimate for a conforming certificate; a checker must read the portion it examines. In particular its time has an $\Omega(Z)$ input-length cost on accepted streams. A short list of J outputs supplies no bound on V and is not a completion proof. These are bounds in the traversal and certificate sizes, not a polynomial-delay assertion in the number of negative outputs.

For fixed v, (41) gives $V \le vC^{v+1}$. The resulting bound is polynomial in numeric C, with degree depending on v. With unary C this is an XP-style upper bound parameterized by v; for binary C it is pseudopolynomial in C. It is not an XP claim in input length $\log C$.

Corollary 21 (Bounded-cardinality decision)**.** *For $1 \le u \le v$, deciding whether every numerical semigroup with $u \le n \le v$ satisfies Wilf's inequality has a uniform deterministic upper bound*

$$2^{O(v \log(v+1))}.$$

The same bound holds for checking the complete finite forest used by this procedure and for the length of one such forest encoding. Working space excluding retained output is $O(v^3 + v^2 \log(v+1))$ bits.

Proof. If $v \le 4$, Theorem 2 supplies the answer. For $v \ge 5$, use $C = H_v = (v-3)(v+1) = O(v^2)$. Corollary 9 confines every possible strict violation to this conductor region. A violation has $W_0 \le W < 0$, so the exact traversal and checker of Theorems 3 and 4 find it among the outputs. If none exists, the complete forest and the cutoff prove the universal statement. Since $V \le vC^{v+1}$ and $J \le V$, substituting into (42)–(44) gives $v^{O(v)}$; the space bound follows from $h \le v$. The binary input length is $O(\log(v+1))$ because $u \le v$, and fits these bounds. □

This corollary neither asserts an affirmative answer for every v nor polynomial time in its binary length. It proves no lower bound or practical feasibility result. The conductor cutoff applies to Wilf violations; it does not truncate the task of listing *all* W_0-negative semigroups. For all left cardinalities at conductor at most C, the separate endpoint is $\lfloor C/2 \rfloor$, as proved in Corollary 16.

6.4 Why the structural parameters matter

At a seed of multiplicity m, $k = q = \lceil c/m \rceil$. Each added gap increases both k and ℓ_P, so an admitted node satisfies $\ell_P \le v - q + 1$. Thus $q > v$ eliminates a root, and otherwise $m \ge \lceil c/v \rceil$ is necessary. The product gate removes all proper descendants once $(k+1)(\ell_P + 1) \ge c$. The large-tail gate uses

the increase of at least $k + \ell_P + 1 - q$ in W_0 along its suffix. These are algebraic explanations of reduced work, without an assumed average branching factor.

The profile method has a different advantage: it can discard many semigroups at once before reconstructing any of them. Its costs depend on the weighted surplus $n - q$, left primitive width s, degree bounds on selected monomials, and the widths of primitive boxes. The detailed counts in Appendix D give an explicit $2^{O(n^2)}$ upper bound for one finite traversal of its relaxed domains. That traversal may return unresolved leaves; the count is not a theorem that every leaf closes. The smaller bound for the canonical method and the profile bound concern different state and output contracts and do not establish which implementation is faster in practice.

7 Numerical shadows and a monomial order ideal

We use Macaulay's theorem for ordinary total degree in a monomial quotient. The shifted Apéry levels locate numerical values, but do not grade that quotient. Since numerical factorizations need not be unique, we distinguish a bound valid for every representation from one obtained by a single choice of representatives.

Lemma 22 (The complete growth convention)**.** *For $j \geq 1$ and $a \geq 0$, write the unique full j-binomial expansion*

$$a = \sum_{i=1}^{j} \binom{a_i}{i}, \qquad a_j > \cdots > a_1 \geq 0, \qquad a_i \geq i - 1.$$

Zero tails have $a_i = i - 1$. Define

$$G_j(a) = \sum_{i=1}^{j} \binom{a_i + 1}{i + 1}, \qquad H_{d,t} = G_{t-1} \circ \cdots \circ G_d \quad (1 \leq d < t). \tag{45}$$

These maps are increasing, $G_j(0) = 0$, and $G_j(a) = a$ for $0 \leq a \leq j$. For the Hilbert dimensions of a standard graded commutative $\mathbb{Q}$-algebra, $h_t \leq H_{d,t}(h_d)$.

Proof. Greedy choice of a_j leaves a remainder strictly below $\binom{a_j}{j-1}$ by Pascal's identity. Continuing downward, with the prescribed zero tails, proves existence and uniqueness. For $1 \leq a \leq j$ there are a leading unit binomials, each still one after growth. For monotonicity, use the successor operation: choose the smallest r with $a_r + 1 < a_{r+1}$, taking $a_{j+1} = \infty$, increase a_r by one, and reset the lower indices to $i - 1$. The preceding indices are consecutive. The hockey-stick identity shows that the original sum increases by one and the grown sum by $a_r - r + 2 \geq 1$. It also covers the zero-tail endpoint. Hence G_j and its compositions are increasing.

Macaulay's theorem [4, Theorem 5.5], in the convention of its Definitions 5.1–5.2 and Notation 5.4, gives $h_{j+1} \leq G_j(h_j)$ in positive degrees; see also [11, §1] for the binomial convention. If $h_j = 0$, generation in degree one gives $R_{j+1} = R_1 R_j = 0$. Iteration proves the assertion, without invoking a degree-zero bound. □

Lemma 23 (Universal numerical shadow)**.** *Let B_0 be a finite set of distinct integers, $1 \leq d < t$, and $K_0 \subset dB_0$ have size a. Let $V \subset tB_0$ have the following property: every degree-t representation of every $v \in V$ has every degree-d partial sum in K_0. Then $|V| \leq H_{d,t}(a)$.*

Proof. The empty-B_0 case is immediate. Enumerate $B_0 = \{b_0, \ldots, b_{s-1}\}$, attach variables X_i, and write $v(X^u) = \sum u_i b_i$. In degrees at most t the additive rank $w(u) = \sum u_i (t+1)^i$ is injective,

since all exponents are at most t. Choose a least-rank degree-d representative for each value of K_0, obtaining a monomials Z_d. Similarly choose least-rank degree-t representatives of V. A degree-d divisor of a chosen degree-t monomial has value in K_0. If it were not the chosen representative of that value, replacing it by the lower-rank representative would keep total degree and numerical value and reduce rank, a contradiction.

In $\mathbb{Q}[X_0, \dots, X_{s-1}]$ kill every degree-d monomial outside Z_d. This is a proper homogeneous monomial ideal. The quotient has degree-d dimension a, and a degree-t monomial survives precisely when all of its degree-d divisors survive. The distinct representatives of V survive, so Macaulay's theorem gives the bound. The hypothesis on every representation is needed when a divisor is replaced by an equal-value representative. □

Lemma 24 (Prefix numerical bounds)**.** *For $2 \le Q \le q$, let B_Q be the primitive left values of level less than Q. If $B_Q = \varnothing$, there is no decomposable Apéry value through level Q. Otherwise let ℓ be its smallest occupied level, and for $2 \le d < t \le Q$ put*

$$L_d = \max\{\ell + d - 1, d(\ell - 1) + 1\}, \qquad a_{Q,t,d} = \sum_{L_d \le i \le Q-t+d} \kappa_i.$$

Then

$$|D \cap \bigcup_{i \le Q} A_i| \le P_Q(\beta) + \sum_{t=3}^{Q} \min_{2 \le d < t} H_{d,t}(a_{Q,t,d}). \tag{46}$$

For $Q < q$ the left side is $\sum_{i \le Q} \kappa_i$; for $Q = q$ it is δ.

Proof. All primitive factorizations through Q use B_Q and have length at most Q. Pair targets are bounded by $P_Q(\beta)$. For a fixed $t \ge 3$, every degree-d partial sum of every representation is proper, left and decomposable; Lemmas 1 and 2 place it between levels L_d and $Q - t + d$. Its numerical image has size at most $a_{Q,t,d}$, so Lemma 23 and monotonicity give the stated shadow bound. Minimize over d and sum over lengths. A value with several lengths may be counted more than once, which only enlarges the upper bound. □

Lemma 25 (All-bipartition occupied-level recursion)**.** *For a multiset l of primitive levels, a singleton (i) is available only when $\beta_i > 0$, and then $\mu((i)) = i$. For length $t \ge 2$, examine every bipartition $v + w = l$ into nonempty submultisets. If a required proper piece is unavailable, the pattern is unavailable. Otherwise set*

$$a(l) = \max_{v+w=l} \{\mu(v) + \mu(w) - 1,\ \mu(v) + 1,\ \mu(w) + 1\}, \tag{47}$$

$$b(l) = \max_{v+w=l} \{\mu(v) + \mu(w),\ \mu(v) + 1,\ \mu(w) + 1\}. \tag{48}$$

For a proper nonsingleton sum define $\mu(l)$ to be the least occupied left level j, meaning $\kappa_j > 0$, such that

$$a(l) \le j \le \min\{q - 1, \textstyle\sum l_i + t - 1\}; \tag{49}$$

if none exists it is unavailable. Every actual proper sum of pattern l has level at least $\mu(l)$. A full target through Q requires $a(l) \le Q$; a noncompressed full target requires $b(l) \le Q$. Full targets are not rounded to left occupancy.

Proof. Induct on the length of l. Every proper piece of an actual representation is Apéry and left, and each bipartition is an actual split. Thus (7) gives every term of $a(l)$. A proper nonsingleton sum lies in R and satisfies (49), so its level is at least the least permitted occupied level; if none

exists, the representation is impossible. For a noncompressed full target, Lemma 5 improves the lower bound for every split to $b(l)$. The full target itself may lie in A_q and is therefore not rounded to a left occupied level. □

Lemma 26 (Formal counts and occupied shadows)**.** *For length $2 \le t \le Q$, enumerate every weakly increasing pattern $1 \le l_i \le Q - t + 1$ satisfying the lower bound in (8) at Q. If level i occurs r_i times, its formal count is*

$$\prod_i \binom{\beta_i + r_i - 1}{r_i}, \tag{50}$$

with factor 1 for $r_i = 0$ and factor 0 for $r_i > 0$, $\beta_i = 0$. Retain available patterns with $a(l) \le Q$, or $b(l) \le Q$ for noncompressed targets. These totals bound the corresponding numerical images. For targets at exact level j, also require $\sum l_i + t - 1 \ge j$ and the lower bound at j.

Let $a_1(Q)$ be the least occupied primitive level below Q, with sentinel Q if no such level exists. For $d \ge 2$ let $a_d(Q)$ be the least available $\mu(l)$ among nonzero-count length-d patterns for a proper target through $Q - 1$. Use sentinel Q if the set is empty. Every proper degree-d subsum of a target through Q has level at least $a_d(Q)$. For $2 \le d < t$, its level lies in

$$a_d(Q) \le j \le \min\{Q - t + d,\ Q - a_{t-d}(Q) + 1\}. \tag{51}$$

For a noncompressed target delete the final $+1$. The formal degree-t cap may therefore be intersected with $H_{d,t}(\sum_j \kappa_j)$ over this range, for every d. Summing the per-length minima gives valid ordinary and noncompressed caps, denoted F_Q, N_Q.

Proof. Lemma 2 contains every actual pattern in the enumerated domain, and stars and bars gives (50). Choosing one monomial per numerical value and applying Lemma 25 proves the formal caps. A proper d-subsum lies below level Q and so has level at least $a_d(Q)$; its complementary $(t-d)$-subsum has level at least $a_{t-d}(Q)$. The floor inequality and the strict increments give (51), with the improved endpoint under noncompression. These bounds hold for every representation and every choice of d factors, as required by Lemma 23. An empty range or unavailable proper sum yields no target. Taking minima and then the union bound gives F_Q, N_Q. □

Lemma 27 (Forced compressed pair subtraction)**.** *Suppose a collection J of left layers is pair-only: for every $j \in J$, every primitive pattern of length $3 \le t \le j$ fails valid exact-level filters. Let p_j count all formal primitive pairs capable of a noncompressed target at level j, and put $f_j = \max(0, \kappa_j - p_j)$. Then the number of noncompressed D-values having a primitive pair representation is at most*

$$U_q(\beta) - \sum_{j \in J} f_j. \tag{52}$$

A negative right side proves the profile impossible.

Proof. Lengths greater than j are impossible, and the hypotheses exclude all other lengths except two. Thus each R_j-value has a primitive pair representation, whose factor levels sum to at most $j + 1 \le q$. It is counted by $U_q(\beta)$. At most p_j of these values can be noncompressed. Choose f_j compressed values and a pair for each. Different values, including values in different layers, give distinct pairs. None can also represent a noncompressed value. Representatives of the noncompressed pair-values inject into the remaining pairs, giving (52). If there are too few pairs even to remove, the required compressed values cannot exist. □

Eliahou's divset construction [6, Proposition 2.15] selects primitive factorizations of minimum length and then lexicographically, obtaining divisor closure. A least-rank replacement argument for quadratic and cubic numerical sums also appears in [12, Lemma C.3, proof]. Here we use minimum degree followed by an additive rank on the left primitive set B, producing one selection across all lengths.

Lemma 28 (A single global order ideal)**.** *Enumerate $B = \{b_0, \dots, b_{s-1}\}$. For every B-representable Apéry value, including zero, select the monomial minimizing lexicographically*

$$(|u|, w(u)), \qquad w(u) = \sum_i u_i (q+1)^i. \tag{53}$$

The selected set Z is divisor closed and its numerical-value map is injective across all degrees. It consists of 1, the s variables, and one monomial for each value of D. Its complement is a homogeneous monomial ideal, so the quotient of $\mathbb{Q}[X_0, \dots, X_{s-1}]$ has basis Z and is standard graded by total degree. For $2 \le d \le q-1$, $1 \le j < q$, put

$$h[d, j] = \#\{u \in Z : |u| = d,\ v(u) \in R_j\}.$$

Then

$$h[d, j] \ge 0, \qquad h[d, j] = 0 \ (d > j), \qquad \sum_{d=2}^{q-1} h[d, j] = \kappa_j. \tag{54}$$

This same allocation h applies to all degrees, prefixes and shadows.

Proof. Representations have degree at most q, so each represented value has a least pair in (53). If v divides a selected monomial u, its value is Apéry by Lemma 1. Replacing v by an equal-value representative v' of smaller degree, or of equal degree and smaller rank, would improve the selection of u. Thus v is selected. The complement of this divisor-closed set is a homogeneous monomial ideal. Its quotient has degree-zero part $\mathbb{Q}$ and is generated in degree one.

There is one representative per value. All of D is B-representable by Lemma 1, whereas a right primitive is not. A left value at level j has degree at most j; hence (54). If $B = \varnothing$, then the quotient is $\mathbb{Q}$, $Z = \{1\}$ and $D = \varnothing$. □

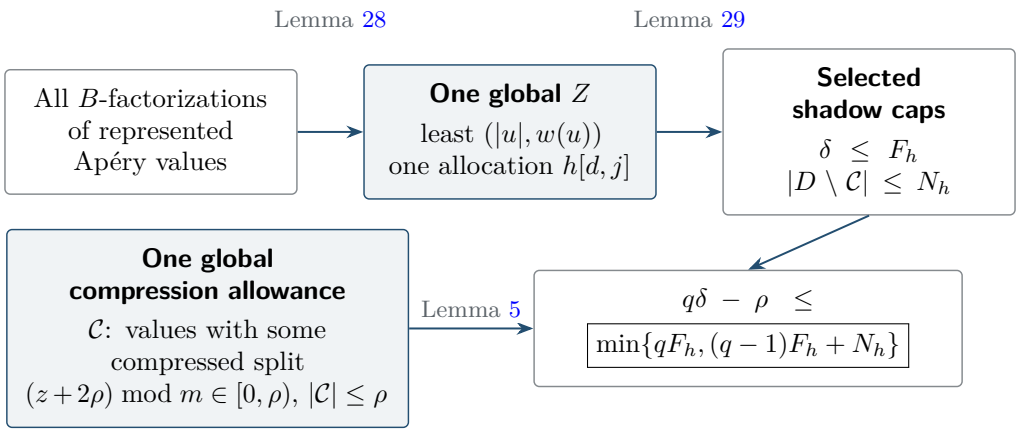


Figure 6. Two shared objects with different definitions. The least-representative selection produces one divisor-closed Z and one h for all selected-shadow bounds. Compression is a property of numerical values, defined using all splits, and its union occupies at most one arc of ρ residues. The displayed bound is one consequence of Lemma 5; its additional U bound and other valid minima may sharpen it. Figures 7 and 2 give the corresponding arithmetic example.

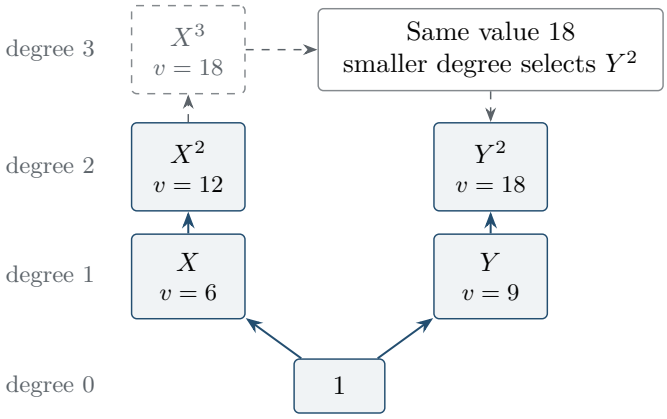


Figure 7. An actual single order ideal for $S = \langle 5, 6, 9\rangle$, $c = 14$, $A = \{0, 6, 9, 12, 18\}$, with $X \mapsto 6$ and $Y \mapsto 9$. The selected monomials are $\{1, X, Y, X^2, Y^2\}$. Although X^3 also has value 18, Y^2 is selected by smaller degree. Solid arrows show divisors of selected monomials; the dashed monomial is excluded. The same selection serves every degree and numerical value.

Lemma 29 (Selected shadows and the exact top split)**.** *Let Y be a subset of selected degree-t targets through $Q \le q$. For every $2 \le d < t$, set $a = \sum_j h[d,j]$ over (51), with its improved endpoint when Y is noncompressed. Then $|Y| \le H_{d,t}(a)$. Intersect these bounds with the degree-t formal counts and any independently proved degree-t cap; call the resulting bounds $C_{Q,t}(h)$ and $C^{\mathrm{nc}}_{Q,t}(h)$. For $t = 2$ there is no shadow minimum. Then*

$$\sum_{j \le Q} h[t,j] \le C_{Q,t}(h) \qquad (Q < q), \tag{55}$$

$$\delta \le F_h := \sum_{t=2}^{q} C_{q,t}(h), \tag{56}$$

$$|D \setminus \mathcal{C}| \le N_h := \sum_{t=2}^{q} C^{\mathrm{nc}}_{q,t}(h).$$

Writing $L_t = \sum_{j<q} h[t,j]$, with $L_q = 0$, the cap $C_{q,t}$ may also be intersected with L_t plus the formal degree-t count at exact level q. For $C^{\mathrm{nc}}_{q,t}$ use all L_t plus only the noncompressed exact top count. Lemma 27 may further bound its degree-two term.

Proof. Every degree-d divisor of a selected target is selected by Lemma 28, proper and left by Lemma 1, and lies in the stated range by Lemma 26. If Y is nonempty, its downward closure is a finite monomial order ideal with degree-t part exactly Y and degree-d part of size at most a. Its standard graded quotient meets Lemma 22, proving the bound. The empty case is immediate. Here divisors of the chosen representatives suffice; Lemma 23 instead required every numerical representation.

The same h applies throughout. Selected values in distinct degrees are disjoint, giving the sums in (56). Nonleft selected values have exact level q, which gives the top split. Counting every L_t left value in the noncompressed cap is conservative, while no forced-left value is added again at the top. Lemma 5 then uses one allowance ρ for F_h, N_h. □

8 Further bounds on Apéry profiles

We sharpen the pair and shadow estimates when the conductor exceeds the known range or the depth is four. Each inequality below is applied only under its stated hypotheses.

Lemma 30 (Cutoff-conditioned patterns and termination)**.** *Suppose a strict violation remains after a conductor-H theorem, so $c \geq H+1$, and suppose $m \leq M_{\max}$ with $M_{\max} > 0$. Then*

$$\frac{\rho}{m} \leq q - \frac{H+1}{M_{\max}}. \tag{57}$$

For a primitive pattern of length t, level sum L, and r factors whose levels are at least two, a target through Q requires

$$(L-Q-1)m < (r-1)\rho. \tag{58}$$

Thus $r \leq 1$, $L \geq Q+1$ is impossible. For $r \geq 2$ the pattern is impossible whenever

$$(L-Q-1)M_{\max} \geq (r-1)\big(qM_{\max} - (H+1)\big). \tag{59}$$

These filters apply to pairs, higher degrees, prefixes and exact layers. Starting from valid caps, repeatedly replace F, N, σ, E by the minimum of the old and newly justified bounds, with $M_{\max} = F+E$. This process terminates at a fixed point or a proved rejection and preserves every strict violation.

Proof. The conductor inequality $qm - \rho \geq H+1$ gives $\rho/m \leq q - (H+1)/m \leq q - (H+1)/M_{\max}$. A level-one primitive is at least $m+1$; the weaker bound m suffices here. A primitive of level $i \geq 2$ is at least $im - \rho$. Thus the pattern sum is at least $Lm - r\rho$, whereas a target through Q is less than $(Q+1)m - \rho$. This proves (58), which is impossible when $r \leq 1$ and $L \geq Q+1$. For $r \geq 2$, division by m and (57) give (59). At known m, ρ, the sharper strict inequality may be used directly.

For a strict violation, $m = e + \delta \leq E + F$. Lemmas 5 and 6 therefore permit updated values of σ, E while retaining all previous bounds. An $E < e_{\min}$ or negative-cap outcome is a contradiction. Otherwise the nonnegative integer caps decrease until stationary.

For clarity, the invariant for the same putative violator before and after each update is

$$\begin{gathered} \delta \leq F, \quad |D \setminus \mathcal{C}| \leq N, \quad q\delta - \rho \leq \sigma, \\ e \leq E, \quad m = e + \delta \leq F + E, \quad c \geq H+1. \end{gathered} \tag{60}$$

Each new pattern bound follows from the old invariant, so taking minima preserves it. Once the stated contradictions are excluded, $F + N + \sigma + E$ is a nonnegative integer and decreases at every nonstationary step. At equality of all four caps the case remains; stationarity itself gives no exclusion. □

Lemma 31 (Top-left and cubic estimates)**.** *The following bounds hold.*

1. *If $q \geq 4$ and $R \subseteq A_{q-1}$, every primitive Apéry factorization has length at most three, and every cubic factor lies in $B_1 \cup B_2$, where $B_i = B \cap A_i$. Consequently*

$$\delta \leq P_q(\beta) + \binom{\beta_1 + \beta_2 + 2}{3}. \tag{61}$$

2. *At actual depth $q \geq 5$, if $\kappa_2 = 0$, the numerical primitive cubic targets through $Q = 5$ number at most*

$$\binom{\beta_1 + \beta_2 + 2}{3} + \kappa_3\beta_3. \tag{62}$$

3. *At actual depth $q \geq 5$, without assuming $\kappa_2 = 0$, their noncompressed subfamily numbers at most*

$$\binom{\beta_1 + 2}{3} + (\kappa_2 + \kappa_3)\beta_2 + \kappa_2\beta_3. \tag{63}$$

The last two statements also apply to level-five prefixes when $q > 5$.

Proof. For the first assertion, a pair in a representation of length at least four is proper left decomposable, hence at level $q - 1$; adding two more factors forces the target above level q. In a cubic, a factor of level at least three and its complementary pair of level $q - 1$ likewise force the target to level at least $q + 1$. Count pairs and the remaining formal cubics to obtain (61). If R is empty, cubics cannot occur, and the bound remains valid.

For the second assertion, every proper pair is at level at least three. A factor of level at least four would force the cubic to level at least six. Two level-three factors are impossible: the pair complementary to one would have level at least four. A cubic with one level-three factor has its complementary pair at exactly level three, so its value belongs to $R_3 + B_3$, with at most $\kappa_3\beta_3$ possibilities. All remaining cubics use only $B_1 \cup B_2$.

For the third assertion, proper pairs have level at least two. Noncompression excludes a factor of level at least four. If there is a level-three factor, its complement is in R_2; if there is a level-two factor, its complement is in $R_2 \cup R_3$. Otherwise all factors lie in B_1. Counting these numerical images gives (63); overlaps can only lower their union's size. □

Lemma 32 (Small shadows and marked quartics)**.** *If at most k quadratics are allowed, for $k = 0, 1, 2, 3, 4$, the cubics all of whose quadratic divisors are allowed number at most $0, 1, 2, 4, 5$, respectively. At actual depth $q \geq 5$, suppose $\beta_1 = 1$ and $\kappa_2 + \kappa_3 \leq 3$. Numerical quartic targets through level five number at most five. The bound is four if no such target uses four B_2 factors, and three if, in addition, none uses one B_1 factor and three B_2 factors. For noncompressed quartics both exclusions hold, so their bound is three.*

Proof. Represent quadratics as loops and ordinary edges. A cube x^3 requires the loop x^2; a cubic x^2y requires that loop and xy; a squarefree cubic requires the three edges of a triangle. With ℓ loops, the count is ℓ plus incidences of ordinary edges at loop vertices plus triangles. For $k \leq 2$ this

is at most k. For $k=3$, the maxima for $\ell=0,1,2,3$ are $1,3,4,3$. For $k=4$, no-loop graphs have at most one triangle, since two require five edges. One loop gives at most four: three incidences form a star and leave no triangle, whereas a triangle leaves at most two incidences. Two loops allow at most three incidences from two distinct edges and no triangle, giving five; three loops give at most $3+2=5$; four loops give four. Lemma 23's injection transfers these formal bounds to the numerical setting when needed.

The extremal enumeration can be checked from the following table. An entry fixes the total number k of allowed quadratic monomials and the number ℓ of loops; a dash means $\ell > k$. The remaining $k-\ell$ edges may be placed on any finite vertex set, but only vertices incident with an allowed edge or loop can occur in a cubic.

k	$\ell=0$	1	2	3	4	maximum
0	0	–	–	–	–	0
1	0	1	–	–	–	1
2	0	2	2	–	–	2
3	1	3	4	3	–	4
4	1	4	5	5	4	5

For example, at $(k,\ell)=(3,2)$ one edge joining the two loop vertices gives two pure cubes and two mixed cubes, attaining four; all other placements give no more. The formula "loops plus incidences at loop vertices plus triangles" checks every table entry.

For quartics through five, strict increments put all factors in $B_1\cup B_2$ and all quadratic subsums in $R_2\cup R_3$. Use the least-rank injection of Lemma 23; at most three quadratics are allowed. Exponent types $2+1+1$ and $1+1+1+1$ require four and six different quadratic divisors and are excluded. A fourth power requires one loop; x^3y requires a loop and an edge; x^2y^2 requires two loops and their edge. With one loop and at most two edges there are at most three quartics. With two loops and one edge there are at most five, with five only when the edge joins the loop vertices: two fourth powers, two $3+1$ monomials and one $2+2$ monomial. Three loops give three. In the five-monomial case at most one of the variables is the unique B_1 variable. If both are in B_2, the first exclusion removes all five. Otherwise it removes the high fourth power, and the second removes the high-cubed, low-single monomial, leaving respectively four and three.

Finally, four high factors split into two pairs each of level at least three. One low and three high factors split into a low-high pair and a high-high pair, also each of level at least three. A noncompressed target would then have level at least six. This proves both exclusions. □

Lemma 33 (Depth-four bounds)**.** *Suppose $q=4$, $\alpha=(a,b,d)$, $\kappa=(0,h,k)$, and $r=a+b$. Put*

$$M=\binom{r+1}{2}+rd,\quad U=\binom{r+1}{2}+ad,\quad P=\binom{r-h+1}{2}+(r-h)(d-k).$$

Then $h\le\min\{b,\binom{a+1}{2}\}$ and

$$\delta\le\min\{M-hd,\ P+G_2(h+k)+H_{2,4}(h)\}. \tag{64}$$

The quartic term may also be intersected with $H_{3,4}(k)$. If $h=0$ and $k=d$, let

$$P_0=\binom{a+b+1}{2},\quad F_0=P_0+G_2(k),\quad V_0=(P_0-k)+\binom{a+1}{2}+ab+\binom{a+2}{3}.$$

Then $\delta\le F_0$, $|D\setminus\mathcal{C}|\le V_0$, and, for every valid cap $\delta\le F$,

$$4\delta-\rho\le 3F+V_0. \tag{65}$$

If $P_0 < k$, the profile is impossible.

Proof. Eligible pairs are within levels one and two, or join those levels to three; lower-sum pairs exclude $2+3$. This gives M, U. Every R_2 value is a pair of A_1 elements, giving the bound on h. Lemma 4 gives $M - hd$, and counting primitive pairs gives P. All pair subsums of a cubic lie in $R_2 \cup R_3$, so Lemma 23 gives $G_2(h+k)$. A quartic has all factors in A_1; a factor at level two with three additional factors would reach level five. Its quadratic subsums lie at level two and its cubic subsums at level three, by the strict-increment bounds. The two shadows give $H_{2,4}(h)$ and $H_{3,4}(k)$. All lengths are at most four.

In the special case $h = 0, k = d$, R lies at level three and Lemma 31 excludes quartics. Every R_3 value is a primitive pair: in a hypothetical cubic its proper pair would already be at level three, forcing the target to level four. These k distinct left values consume k of the P_0 pairs, leaving at most $P_0 - k$ distinct top pair values. A noncompressed left pair must have an A_1 factor, since a $2+2$ split at level three is compressed; these values number at most $\binom{a+1}{2} + ab$. A cubic with an A_2 factor has its complementary pair in R_3, making a level-four target compressed. All remaining noncompressed cubics use A_1 only. Their formal count is $\binom{a+2}{3}$. This proves V_0 and hence (65) by Lemma 5. The universal pair shadow proves F_0. □

Lemma 34 (Two further depth-four bounds)**.** *If $q = 4$ and $A_1 = \varnothing$, put $b = \alpha_2$, $d = \alpha_3$, $k = \kappa_3$,*

$$J = A_4 \cap (A_2 + A_2), \quad \ell = |J|, \qquad H_0 = (A_4 \cap D) \setminus (A_2 + A_2), \quad h_0 = |H_0|.$$

Then

$$\begin{aligned} &\delta = k + \ell + h_0, \quad k + \ell \le \binom{b+1}{2}, \quad e \ge 1 + b + d - k, \\ &k + h_0 \le \rho, \quad h_0 \le \min\{bd, b(d-k) + G_2(k)\}, \\ &4\delta - \rho \le 2b(b+1) - k + 3h_0. \end{aligned} \tag{66}$$

If instead every element of $U_0 = A_1 \cup A_2$ is primitive, put $r_0 = |U_0|$, $d = \alpha_3$, $k = \kappa_3$. Then

$$\delta \le \binom{r_0+1}{2} + G_2(k) + r_0(d-k). \tag{67}$$

Proof. In the first case A_2 is primitive. Pairs giving level three use $A_2 + A_2$, and top decompositions use $A_2 + A_2$ or $A_2 + A_3$. This proves the partition, pair count, and lower bound for e. The R_3 values are compressed by $2+2$, and the H_0 values by $2+3$ at level four; they are distinct sets, so one application of Lemma 5 gives $k + h_0 \le \rho$. There are at most bd values in $A_2 + A_3$. Those in $A_2 + (A_3 \setminus R_3)$ number at most $b(d-k)$. Every remaining H_0 value admits an A_2-cubic representation; in every such representation, every pair subsum is proper left and must have level three, because adding a level-two factor reaches at most four. Lemma 23 bounds this cubic image by $G_2(k)$. Finally,

$$\begin{aligned} 4\delta - \rho &\le 3k + 4\ell + 3h_0 \\ &= 4(k+\ell) - k + 3h_0 \le 2b(b+1) - k + 3h_0. \end{aligned}$$

For the second case, eligible decompositions are $U_0 + U_0$, $U_0 + (A_3 \setminus R_3)$, or $U_0 + R_3$. Every R_3 value is a U_0-pair. Every pair subsum in any U_0-cubic Apéry value is left decomposable and hence, by primitivity of U_0, lies in R_3. Lemma 23 gives $G_2(k)$ for that numerical cubic image. Counting the first two images proves (67). □

9 Finite domains and value constraints

The profiles and boxes below are necessary conditions, not realizability tests. Their purpose is to retain the data of every putative strict violation through the final matching bounds.

Lemma 35 (Complete finite domains)**.** *For a fixed positive n, enumerate $q = 1, \dots, n$, disposing of $q = n$ by Lemma 2. For every remaining depth not covered by an imported result enumerate all nonnegative integer solutions*

$$\sum_{i=1}^{q-1} (q-i)\alpha_i = n - q \tag{68}$$

and every rectangle

$$\kappa_1 = 0, \qquad 0 \le \kappa_i \le \alpha_i \quad (2 \le i < q). \tag{69}$$

Set $\beta = \alpha - \kappa$, $M = P_q(\alpha)$, $U = U_q(\alpha)$, initially $F = M$, $\sigma = \min(qF, (q-1)F + U)$, and let E be as in (13). Use every allocation h in (54), subject only to proved necessary coordinate or prefix bounds. Refine caps by the preceding lemmas. Every remaining violation lies in the finite integer domain

$$\begin{gathered} e_{\min} \le e \le E, \qquad K \le \delta \le F, \qquad m = e + \delta, \\ 0 \le \rho < m, \qquad c = qm - \rho > ne, \qquad c \ge H + 1 \text{ when the cutoff is used.} \end{gathered} \tag{70}$$

Every e in this interval is the full embedding dimension.

Proof. Lemma 1 gives the actual depth and profile. The bound $0 \le \alpha_i \le \lfloor (n-q)/(q-i) \rfloor$ makes (68) finite. Choosing each coordinate up to the current remainder divided by its weight, then taking the last coordinate to be that remainder, enumerates every solution. The rectangle contains the actual decomposable counts. Lemmas 3, 5, and 6 bound F, E; $\delta \ge K$ and $m = e + \delta$ give (70). Cited results are used only under their respective hypotheses.

Lemma 28 supplies one actual h. In each layer every bounded composition of κ_j among degrees $2, \dots, j$ is included; coordinate and prefix inequalities remove only impossible compositions. If none remains, the profile is excluded. All refinements preserve the data of the same putative violator. Completeness of the domain alone does not establish that every case has been excluded. □

Lemma 36 (Simultaneous rank boxes)**.** *For an exact tuple, name the β_i primitives in level i in increasing order $x_{i,a}$, $0 \le a < \beta_i$. Then*

$$\begin{gathered} \max\{m+1, im-\rho\} + a \le x_{i,a} \le (i+1)m - \rho - \beta_i + a, \\ x_{i,a+1} \ge x_{i,a} + 1. \end{gathered} \tag{71}$$

Write the resulting flattened box as $L_i \le x_i \le U_i$. A monomial u has numerical interval

$$[\textstyle\sum u_i L_i, \sum u_i U_i].$$

The recursion of Lemma 25 may be strengthened for every proper divisor v by rounding its split lower bound to the least occupied left level j whose inclusive layer intersects v's numerical interval, with $j \le \min(q-1, \sum l_i + |v| - 1)$. If none exists the representation is impossible. At a full target use the ordinary or noncompressed split bound, its interval, and its intended prefix or exact layer, without left rounding of a top target. Counting every named exponent vector that passes these tests gives a sound formal cap.

Proof. A primitive lies in its level interval and is at least $m+1$. The a smaller primitives of the same level reserve a lower positions, and the $\beta_i - 1 - a$ larger ones reserve that many upper positions. These give (71) for the entire ordered value vector.

Every proper divisor of an actual selected representation is left, lies in its box-sum interval, and satisfies each split bound. Induction on degree therefore makes the recursively rounded minimum no greater than its actual level. The full target lies in its intended interval and layer as well. Distinct selected values have distinct surviving named monomials. Ignoring correlations among box coordinates only enlarges the domain. If $B = \varnothing$, there are no variables or decomposable targets. □

Lemma 37 (All-degree left labels)**.** *Fix h, and set*

$$r = \max(\{1\} \cup \{d : \textstyle\sum_j h[d,j] > 0\}).$$

Every selected top monomial has degree at most $\min(q, r+1)$*. Enumerate every monomial of every degree $2, \dots, r$, completing lower degrees first, and label it zero or a left level j. Use exact quotas $h[d,j]$. A positive label must pass the factor-level, occupied-level, rank and interval tests for exact layer j. At degree at least three, every distinct immediate divisor must already have a positive label strictly smaller than j. Label zero means only "not selected left."*

A quota lookahead may exclude a branch only if a layer's remaining quota exceeds the number of remaining eligible monomials, or the total quota exceeds all remaining monomials. Every eligible positive choice and the zero choice whenever quotas can still be met must be included. At a completed labeling, the top-candidate set consists of every monomial of degrees $2, \dots, \min(q, r+1)$ which is not labeled left, has every immediate divisor labeled left when $|u| \geq 3$, passes the necessary level and rank tests, and has nonempty interval

$$I_u = [\sum_i u_i L_i, \sum_i u_i U_i]_{\mathbb{Z}} \cap [qm - \rho, (q+1)m - \rho - 1]_{\mathbb{Z}}. \tag{72}$$

This includes all actual selected top monomials.

Proof. A selected top monomial of degree at least $r+2$ would have a selected left divisor of degree $r+1$, contrary to the definition of r. The actual selection has the required quotas. Its immediate divisors are selected left and have smaller level, and iteration covers all lower divisors. The other tests are necessary by the preceding lemmas, so the complete enumeration retains this labeling. An eligibility set computed before subsequent box tightening may overestimate choices but cannot exclude the actual labeling. At completion every quota is zero.

The divisor and interval properties also put every actual selected top monomial in (72). If $K = 0$, then $r = 1$ and only degree-two top monomials are needed. Repeated variables are allowed, but immediate divisors are distinct monomials rather than positions. The relaxed labels impose no numerical collision condition; zero means only that the monomial is not selected left, and permits top selection. □

Lemma 38 (Multiplicity-sensitive propagation)**.** *A monomial u labeled j imposes $a = jm - \rho \leq \sum u_i x_i \leq b = (j+1)m - \rho - 1$. For each $u_i > 0$, the updates*

$$L_i \leftarrow \max\left\{L_i, \left\lceil \frac{a - \sum_{k \neq i} u_k U_k}{u_i} \right\rceil\right\}, \tag{73}$$

$$U_i \leftarrow \min\left\{U_i, \left\lfloor \frac{b - \sum_{k \neq i} u_k L_k}{u_i} \right\rfloor\right\} \tag{74}$$

preserve all solutions. So do the adjacent same-level updates $L_{\text{next}} \geq L_{\text{previous}} + 1$ *and* $U_{\text{previous}} \leq U_{\text{next}} - 1$*. Repeated complete passes terminate at a fixed point or an empty coordinate. Only emptiness is a contradiction.*

Proof. In any solution the other terms lie between their current lower and upper sums. Isolating $u_i x_i$ gives (73) with exact integer ceiling and floor, including for negative numerators. Each update and each order constraint intersects an old interval with a necessary bound. Lower endpoints increase and upper endpoints decrease within finite initial intervals, so repeated passes terminate. The denominator is the exponent u_i; repeated factors are not independent coordinates. A nonempty resulting box need not be realizable. □

Lemma 39 (Joint integer maximum matching)**.** *Build a bipartite graph whose left vertices are all top candidates of all degrees, whose right vertices are the integers in the top layer, and whose edges join* u *to every integer of* I_u*. If* $T_{\max}$ *is its maximum matching cardinality, then*

$$\delta \leq K + T_{\max}. \tag{75}$$

Its exact maximum is computed by sorting the nonempty intervals by nondecreasing upper endpoint and assigning each the smallest unused integer it contains, skipping it only when none remains.

Proof. Actual selected top values are numerically distinct even across degrees, by Lemma 28. Their monomial-value pairs form a matching in this one joint graph. There are $\delta - K$ such values, proving (75).

For the algorithm, inductively take a maximum matching agreeing with all preceding greedy decisions. If the current interval has no unused point, all its points are occupied by those fixed decisions and it must be skipped. Otherwise let x be its smallest unused point. If the maximum matches this interval to y, then $y \geq x$. If x is free, replace y by x. If x is assigned to a later interval J, swap y to J: its lower endpoint is at most $x \leq y$, and its upper endpoint is at least the current upper endpoint, hence at least y. If the current interval is unmatched, transfer x from its later owner to it, preserving cardinality; if x too were free, adding it would contradict maximality. Thus a maximum can agree with the next greedy decision, proving the claim by induction. Equal endpoints and repeated intervals cause no change. Merely finding some matching would give a lower bound and could not justify an exclusion. □

Lemma 40 (Joint residue maximum matching)**.** *Take left vertices for zero, the* s *primitive variables, the* K *positively labeled left monomials, and all optional top candidates of all degrees. Their numerical intervals are respectively* $\{0\}$*,* $[L_i, U_i]$*, the monomial box-sum interval intersected with its labeled layer, and* I_u*. Join each vertex to every residue modulo* m *attained by its interval. If* $R_{\max}$ *is the unrestricted maximum matching cardinality, then*

$$1 + s + \delta \leq R_{\max}, \qquad \delta \leq R_{\max} - 1 - s. \tag{76}$$

Right primitive vertices may be omitted. No mandatory-coverage constraint is needed when computing this maximum.

Proof. The actual subset $\{0\} \cup B \cup R \cup (D \cap A_q)$ has $1 + s + \delta$ elements of A, all in distinct residues. Their values lie in the stated intervals and give a matching using the mandatory vertices and the appropriate top candidates. Its size cannot exceed the unrestricted maximum. Right primitives are absent from this matching; their omission does not set $e = 1 + s$.

An inclusive interval with at least m integers contains all residues. For shorter intervals enumerate its integers modulo m, including wraparound; this gives exactly its residue set. To find the maximum,

start empty and add left vertices one at a time. Each new vertex can increase the optimum by at most one. If a larger matching exists, the symmetric difference with the current one contains an augmenting path starting at the new vertex: a path entirely in the old prefix would contradict its maximality. Exhaustive alternating search follows owners of matched residues, marking visited residues to prevent cycles. Reaching a free residue permits a flip; failure to find one proves no augmenting path exists. Induction proves exactness. Again, a nonmaximum matching would be insufficient. □

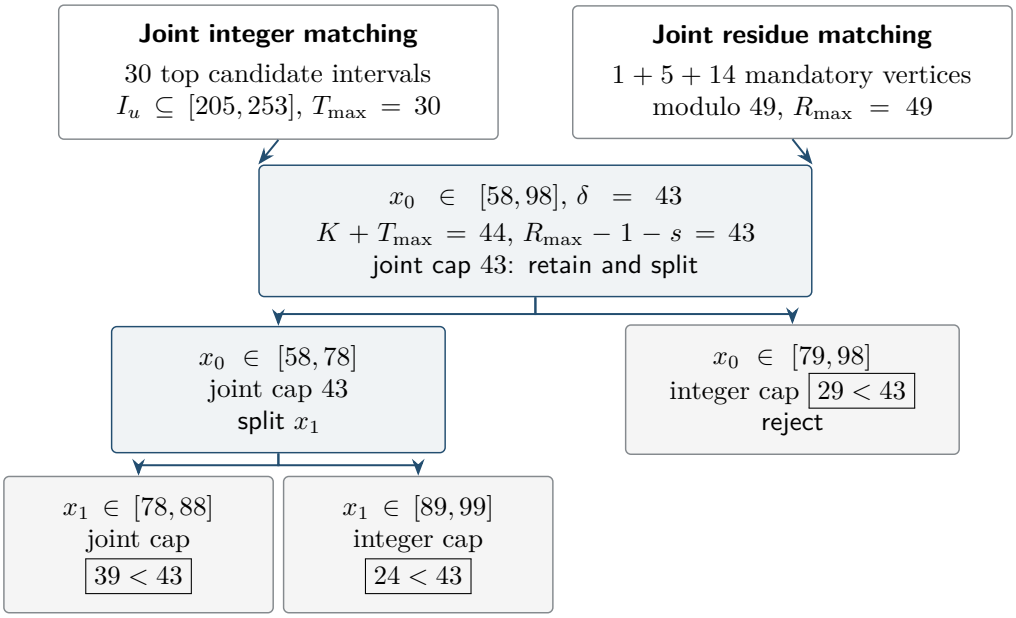


Figure 8. Packing for one completed labeling at $n = 34$. Integer and residue matchings use all candidate degrees in their respective graphs. The displayed children partition the parent coordinate; propagation and packing bounds close each child. The diagram closes this labeling by partitioning each surviving box.

Lemma 41 (Covering splits and unresolved leaves)**.** *At a completed labeling, a box may be rejected by an empty coordinate or by a proved necessary bound such as*

$$\min\{K + T_{\max}, R_{\max} - 1 - s\} < \delta = m - e. \tag{77}$$

Otherwise choose a coordinate with $L_i < U_i$, put $z = \lfloor (L_i + U_i)/2 \rfloor$, and visit both children replacing its interval by $[L_i, z]$ and $[z + 1, U_i]$. Recompute propagation and candidate bounds in both. If every terminal leaf of this finite tree is contradicted, the original box is excluded. A passing singleton, resource stop, omitted child, or any noncontradicted leaf is unresolved.

Proof. The children partition the parent integer box. An actual vector lies in one child and remains there under Lemma 38's propagation; Lemmas 37–40 retain its candidates and matchings. Induction from contradicted leaves therefore excludes a closed tree. Each split shortens a finite integer interval, but a passing singleton is still unresolved. The same applies to a zero-coordinate box whose necessary tests do not contradict it, and to a computation stopped before closure. □

Proposition 42 (Finite-cover soundness)**.** *Fix a finite set of positive cardinalities. Suppose a finite computation covers all domains of Lemma 35, using one shared h on each branch, all labels of*

Lemma 37, and both children of every split in Lemma 41. Suppose every discarded family has a specific applicable imported result or necessary contradiction proved above, and every remaining terminal branch is contradicted. Then Wilf's inequality holds at every cardinality in that set, relative to the imported results actually used. No omitted branch or unresolved leaf is permitted in this premise.

Proof. Suppose S were a strict violation in the stated cardinalities. None of the applicable cited results excludes it. Lemmas 1 and 2 give its $q < n, \alpha, \kappa, \beta$, and Lemmas 3–6 place its full e, δ, m, ρ in the finite domain. Every subsequent refinement retains these values. Lemma 28 gives one order ideal and one enumerated allocation h for all its shadows.

Lemma 36 contains its ordered primitive value vector, and Lemma 37 includes its left labeling with exact quotas and divisor conditions. Propagation retains the vector; every split places it in exactly one child. Its selected top values give matchings in both joint graphs, so neither maximum can exclude it. The complete cover would therefore lead this vector to a contradictory leaf, impossible. Hence no strict violation exists. The proposition requires the stated closure of the finite domain. □

10 A uniform application at depth six

The selection of Lemma 28 also gives structural results without a bound on the number of left elements. We illustrate this with a family whose non-m left primitives occupy two consecutive ordinary intervals. The selected quadratic and cubic monomials suffice to prove strict positivity of the Eliahou number. Thus the same order ideal used for the simultaneous shadow estimates has a uniform application.

In this section write $r = r_P = |P \cap [c, c+m)|$ for the number of right primitives. Every primitive is less than $c+m$, so $|P \cap L| = e - r$ and there are $m - r$ decomposable elements in $[c, c+m)$. In the present notation, Eliahou's quantity in the published version [4, Definition 3.10 and Proposition 3.11] is

$$W_0(S) = (e-r)n - q(m-r) + \rho, \qquad W(S) = W_0(S) + r(n-q). \tag{78}$$

We write W_0 for the Eliahou number to distinguish it from the embedding-dimension cap E in (13).

Theorem 6 (A mixed-interval family)**.** *Let S be a numerical semigroup of multiplicity m and conductor $c = 6m - \rho$, where $0 < \rho < m$. Suppose its primitives below c are exactly*

$$P \cap L = \{m\} \sqcup \{2m + x : x \in X\} \sqcup \{3m + y\}, \tag{79}$$

where $\varnothing \neq X \subseteq \{1, \dots, m-1\}$ and $y \in \{1, \dots, m-1\} \setminus X$. Put

$$k = |X|, \qquad \ell = \#\{z \in X \cup \{y\} : z < m - \rho\}.$$

Then, with no restriction on the right primitives,

$$W_0(S) = (k+2)n - 6(m-r) + \rho \;\geq\; \begin{cases} \rho, & \ell \geq 1, \\ \rho - k + 4 \geq 5, & \ell = 0. \end{cases} \tag{80}$$

In particular, $W_0(S) > 0$.

The intervals in (79) are ordinary numerical intervals, rather than the shifted layers A_i. For example, $\lambda(2m + x)$ may be either 2 or 3. The proof uses genuine monomial degree and does not assume that shifted numerical levels are additive.

Proof. Here $q = 6$, $e = k + 2 + r$, and $B = \{2m + x : x \in X\} \sqcup \{3m + y\}$. Use the existing selection Z of Lemma 28, on the $k + 1$ variables corresponding to B. By Lemma 1, every decomposable Apéry value is represented using B. A right primitive is an Apéry value but is not B-representable. Consequently the selected values are exactly $\{0\} \sqcup B \sqcup D$, and

$$|Z| = m - r. \tag{81}$$

This uses one global selection; right primitives contribute no variables to Z. Divisor-closed Apéry models of this kind are also constructed in [6, Proposition 2.15].

Call a variable of value $2m + x$ a type-2 variable, and call the variable of value $3m + y$ the type-3 variable. Every Apéry value is at most $c + m - 1 = 7m - \rho - 1$. A product of four variables has value at least $8m + 4$, and a product of three variables containing the type-3 variable has value at least $7m + 3$. Neither can be selected. Thus the only composite types in Z are 22, 23, 33 and 222. Denote their counts by $p_{22}, p_{23}, p_{33}, h_3$, respectively. Since Z contains the unit and all $k + 1$ singleton variables, (81) gives

$$m - r = k + 2 + p_{22} + p_{23} + p_{33} + h_3, \qquad p_{22} \le \binom{k+1}{2}, \quad p_{23} \le k, \quad p_{33} \le 1. \tag{82}$$

These are formal monomial capacities; selected numerical values remain distinct even when different formal products have the same value.

Let t count the selected type-22 pairs whose offset sum is less than $m - \rho$. A selected type-222 monomial has value $6m + x_1 + x_2 + x_3 \le 7m - \rho - 1$, with repetitions allowed. Hence its total offset is less than $m - \rho$, and every quadratic divisor is one of the t pairs. The following weighted count gives the cubic shadow estimate:

$$3h_3 \le (k + 2)t. \tag{83}$$

Indeed, write the exponent vector of a selected cubic in the k type-2 variables as γ. For each i with $\gamma_i > 0$, give its incidence with the quadratic $\gamma - \varepsilon_i$ weight γ_i, where ε_i is the ith coordinate vector. The total weight from each cubic is 3. A fixed quadratic exponent θ can receive weight only from parents $\theta + \varepsilon_i$, and their total weight is at most $\sum_{i=1}^{k}(\theta_i + 1) = k + 2$. Divisor closure proves (83), including the case $t = 0$. The weights are essential for repeated factors: a cube has only one distinct quadratic divisor. This is an elementary form of the classical degree-two to degree-three Hilbert-growth estimate; compare [4, Theorem 5.10, published version].

We next count elements below c by Apéry residue rows. The row of $a \in A$ consists of $a, a + m, a + 2m, \ldots$, and different selected values have different residues. Put $\eta(z) = 1$ if $z < m - \rho$, and $\eta(z) = 0$ otherwise. The following rows supply a lower bound for n:

Selected value	Number of rows	Elements below c per row
0	1	6
$2m + x$, $x \in X$	k	$3 + \eta(x)$
$3m + y$	1	$2 + \eta(y)$
$4m + x + x'$, $x + x' < m - \rho$	t	2

For instance, the row of $2m + x$ always contributes its first three elements; its fourth lies below c exactly when $x < m - \rho$. A low-offset pair contributes both $4m + x + x'$ and $5m + x + x'$. Discarding the other nonnegative row contributions yields

$$n \ge 3k + 8 + \ell + 2t. \tag{84}$$

Substitute (82) and (84) into the definition of W_0. First use the capacities for p_{22} and p_{23}, and then (83), to obtain

$$\begin{aligned} W_0(S) &\geq (k+2)(3k+8+\ell+2t) - 6(k+2+p_{22}+p_{23}+p_{33}+h_3) + \rho \\ &\geq \rho - k + 4 + (k+2)\ell - 6p_{33} + 2(k+2)t - 6h_3 \\ &\geq \rho - k - 2 + (k+2)\ell + 6(1-p_{33}). \end{aligned} \tag{85}$$

If $\ell \geq 1$, the last expression is at least ρ, since $p_{33} \leq 1$. If $\ell = 0$, the $k+1$ distinct offsets in $X \cup \{y\}$ all lie in the ρ integer positions $[m-\rho, m-1]$, so $\rho \geq k+1$. Moreover a selected type-33 pair would have value $6m + 2y \leq 7m - \rho - 1$, forcing $2y < m - \rho$. This is impossible when $y \geq m - \rho > 0$. Thus $p_{33} = 0$, and (85) gives $W_0(S) \geq \rho - k + 4 \geq 5$, as claimed. □

Corollary 43. *Every semigroup satisfying the hypotheses of Theorem 6 satisfies* $en - c > 0$.

Proof. The zero residue row contains $0, m, \ldots, 5m$ below c, so $n \geq 6$. Equation (78) gives $W(S) = W_0(S) + r(n-6) > 0$. □

An example with a right primitive. Consider $S = \langle 8, 18, 21, 31, 51 \rangle$. Its Apéry values, listed in residue order modulo 8, are

$$(0, 49, 18, 51, 36, 21, 54, 31).$$

Their maximum is 54, giving $c = 47 = 6 \cdot 8 - 1$. The displayed generators are minimal, and 51 is the sole right primitive. Thus $X = \{2, 5\}$, $y = 7$, $k = 2$, $r = 1$ and $\ell = 2$. The left set is

$$\{0, 8, 16, 18, 21, 24, 26, 29, 31, 32, 34, 36, 37, 39, 40, 42, 44, 45\},$$

so $n = 18$ and $e = 5$.

Let U_2, U_5, V represent $18, 21, 31$, respectively. The selection is

$$Z = \{1, U_2, U_5, V, U_2^2, U_2V, U_2^3\},$$

with respective values $0, 18, 21, 31, 36, 49, 54$. In particular, $|Z| = 7 = m - r$, $p_{22} = t = 1$, $p_{23} = 1$, $p_{33} = 0$ and $h_3 = 1$. The cube U_2^3 illustrates the weighted incidence count, and (84) is an equality. Directly,

$$W_0(S) = 4 \cdot 18 - 6 \cdot 7 + 1 = 31, \qquad W(S) = 5 \cdot 18 - 47 = 43.$$

The difference is $12 = r(n-6)$, retaining the contribution of the right primitive to the full embedding dimension.

Relation to the finite cover. Theorem 6 uses the same global selection as the finite reduction, but applies at every left-element cardinality compatible with its structural hypotheses. The example lies in the cardinality interval of Theorem 7. The structural theorem is a companion application: the reported traversal retains its stated rules and counts, and its proof does not depend on this additional theorem. The following section presents that recorded finite-cover application.

11 The recorded cardinality application

Theorem 7. *Every numerical semigroup with* $13 \leq |S \cap [0, c)| \leq 37$ *satisfies* $e|S \cap [0, c)| \geq c$.

This application uses the mathematical inputs and external conductor verification listed in Section 1.4, together with the recorded residual computation described below. Its full-range computational evidence and validation limits are retained from the preceding edition; this distribution does not extend the enumerated range. The programs and finite records for this application are distributed in the Parallel Attack computational companion, version `companions-v1`; Appendix C gives the execution archive and verification instructions. The newer general algorithms in Sections 4 and 5 are not claimed to have been executed by this companion.

11.1 The algebraic reduction at thirteen left elements

The initial bounds already close the reduction at $n = 13$. Suppose, indeed, that $c > 13e$. The small-embedding-dimension and small-depth results give $e, q \geq 4$, while the graph consequence excludes $n \leq 3q - 1$. Hence $q = 4$. Write $\alpha = (a, b, d)$. The Apéry identity gives $3a + 2b + d = 9$. Put $r = a + b$, so $0 \leq r \leq 4$ and $d = 9 - 2r - a \leq 9 - 2r$. The eligible-pair bound gives

$$\delta \leq M = \binom{r+1}{2} + rd \leq \frac{r(19 - 3r)}{2} \leq 15.$$

The last inequality follows by evaluating the five integers $r = 0, \ldots, 4$. Consequently

$$13e < c = 4(e + \delta) - \rho \leq 4e + 60.$$

Thus $e \leq 6$ and $c \leq 84$. Bacher's verification for $c \leq 200$ excludes this strict violation.

11.2 Enumeration

We implemented the procedure of Appendix B in Python, using exact integer arithmetic. The reduction applies throughout $13 \leq n \leq 37$. Its initial bounds close $n = 13$ directly (Section 11.1); the recorded runs cover $n = 14, \ldots, 37$. All branches were excluded, with no unresolved leaf or omitted child. There was no node, depth or time limit. The largest splitting depth encountered was ten.

The enumeration follows the bounds in Sections 2, 7, 8 and 9. It visits weighted profiles, decomposable profiles, allocations of selected monomials, exact integer tuples, and labelings of primitive-value boxes. A branch is discarded by an applicable preliminary result or by one of the necessary inequalities proved above. Proposition 42 shows that exclusion of every branch proves Wilf's inequality. Table 4 lists the estimates used at each stage.

11.3 Results

For $14 \leq n \leq 32$, every branch closed during profile refinement. For $33 \leq n \leq 37$, the computation also used global allocations, integer tuples, all-degree labels and the matching tests.

Table 1 gives the profile counts. The α column counts weighted profiles at the residual depths. "Empty" counts exclusions by item 3; "raw" counts exclusions by the initial conductor bound. The κ column counts rectangle points for profiles remaining after these tests, and "survive" counts those reaching allocation. All entries refer to enumerated search states.

Table 1. Executed residual profile domains.

n	α	Empty	Raw	κ	Survive
14	14	6	8	0	0
15	39	12	27	0	0
16	46	13	33	0	0
17	55	14	41	0	0
18	110	34	76	0	0
19	131	37	94	0	0
20	154	41	113	0	0
21	271	69	202	0	0
22	320	76	240	120	0
23	379	83	286	299	0
24	606	155	428	654	0
25	720	171	504	1530	0
26	847	191	582	3010	0
27	1283	294	874	5006	0
28	1513	327	1027	7344	0
29	1786	362	1199	11055	0
30	2577	592	1678	17195	0
31	3044	661	1934	28093	0
32	3573	738	2205	43390	0
33	4996	1075	3046	64896	60
34	5871	1201	3556	89551	173
35	6894	1338	4151	122870	379
36	9355	2019	5483	176079	707
37	10985	2259	6279	248503	1216
Total	55569	11768	34066	819595	2535

Table 2 gives the subsequent stages. “Allocations” counts completed global compositions after coordinate and prefix pruning; “remain” counts those surviving refinement. “Tuples” counts (e, δ, m, ρ, c) after (91), and “roots” counts those reaching packing after ranked caps. “Labels” counts completed labelings; “boxes” counts matching boxes, including children. Every listed singleton was examined and excluded.

n	Allocations	Remain	Tuples	Roots	Labels
33	440	6	78	62	12734
34	1327	19	342	295	156972
35	3521	31	661	560	407977
36	7644	58	864	687	837497
37	13593	105	1469	1122	2626961
Total	26525	219	3414	2726	4042141

n	Boxes	Splits	Singletons	Max. depth
33	12734	0	0	0
34	156990	9	0	2
35	408619	321	1734	10
36	838067	285	4881	7
37	2627241	140	3371	9
Total/max.	4043651	755	9986	10

Table 2. Executed allocation, tuple and packing stages. All entries for $n = 14, \dots, 32$ are zero. The last column is a maximum, not an additive count.

Table 3. A branch of the enumeration at $n = 34$. Event indices are local to each stage.

Recorded event	Evaluated data and implication
Profile 10563	$q = 5$, $\alpha = (0, 5, 0, 14)$, $\kappa = (0, 0, 0, 14)$, $F = 45$, $N = 15$, $\sigma = 195$, $E = 6$, $C = 225$.
Allocation 374; tuple 254	$h[2, 4] = 14$ is the only nonzero allocation. $(e, \delta, m, \rho, c) = (6, 43, 49, 40, 205)$; $205 > 34 \cdot 6 = 204$.
Label 17862; box 17863	Fourteen left pairs have label 4. The box has $T_{\max} = 30$, $R_{\max} = 49$, $K = 14$, $s = 5$; $\min\{K + T_{\max}, R_{\max} - 1 - s\} = 43 = \delta$.
Split 17864	$x_0 \in [58, 98]$ partitions into $[58, 78]$ and $[79, 98]$.
Left 17866–17873	The first child still has cap 43 and splits $x_1 \in [78, 99]$ into $[78, 88]$ and $[89, 99]$. After propagation their joint and integer caps, respectively, are 39 and 24, both below 43.
Right 17874–17876	The other x_0 child has integer cap $29 < 43$ after propagation. This closes both children of split 17864.

In total, there were 55 569 weighted profiles and 819 595 rectangle points. Profile refinement retained 2 535 points. Of 26 525 global allocations, 219 remained after refinement; these gave 3 414 exact tuples and 2 726 packing roots. Quota lookahead rejected 4 231 199 branches and propagation rejected 1 339 441. The packing stage completed 4 042 141 labelings and 755 splits, evaluated 4 043 651 boxes, and obtained 4 042 896 matching contradictions. In particular,

$$4\,042\,141 + 2(755) = 4\,043\,651, \qquad 4\,042\,141 + 755 = 4\,042\,896.$$

The traversal recorded the profile, allocation, tuple and labeling trees, including local exclusions and both children of each split. Validation checked their finite domains, cap arithmetic, coverage and exact matching maxima. The record validator does not regenerate every label-eligibility, rank, propagation or candidate-graph step; a complete second run with the same implementation and scientific kernels reproduced the closed cover. The profile and first refinement stages were additionally reconstructed for $14 \le n \le 32$, all reached stages were checked for $33 \le n \le 36$, and tuple, labeling and packing decisions were checked for $n = 37$. Appendix C treats the two implementation details

relevant to the terminal exclusions. The formulas and pseudocode in Appendix B permit an independent implementation.

11.4 The three ranges of the reduction

The identity $n - q = \sum_{i=1}^{q-1}(q-i)\alpha_i$ restricts the eligible-pair counts particularly strongly at small n. For $13 \le n \le 21$, preliminary exclusions and initial conductor bounds suffice. At $n = 13$, Section 11.1 gives $c \le 84$. For $14 \le n \le 21$, after the early-empty exclusion the largest initial conductor bound grows from 74 to 196. Bacher's reported conductor verification therefore excludes every remaining profile before a κ rectangle is entered. The four $n = 21$ profiles with an unconditioned endpoint above 200 have $q = 4$ and $\alpha_1 = 0$, so they are removed by the early-empty result.

At $n = 22$, an initial endpoint first exceeds 200; its maximum is 215. The computation consequently visits 120 rectangle points at $n = 22$ and 43 390 at $n = 32$. Decomposable-profile and shadow refinements exclude all of them before allocation of h. At $n = 33$, the first refined profiles reach allocation. Their number increases from 60 at $n = 33$ to 1 216 at $n = 37$, where the selected order ideal, integer tuples and joint matching bounds are needed.

Completed labelings increase from 12 734 at $n = 33$ to 2 626 961 at $n = 37$. This reflects both the growth from 62 to 1 122 packing roots and the increase in labels per root, from about 205 to 2 341. Earlier conductor exclusions continue to apply throughout the larger cases. Figure 9 summarizes the three ranges of this reduction.

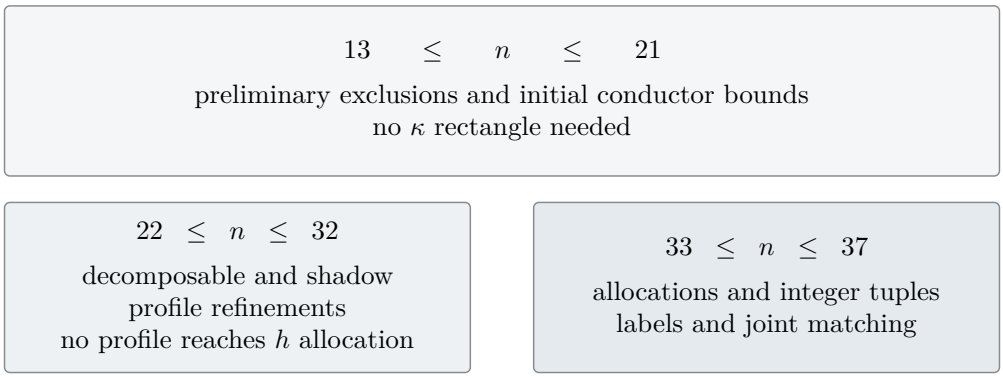


Figure 9. The three ranges of the conductor reduction for $13 \le n \le 37$. The cells give the deepest stage used for the exclusions; later stages also use the preceding bounds. The first range uses the preliminary exclusions before the conductor cutoff. See Section 11.4 and Tables 1–2.

Proof of Theorem 7. Throughout $13 \le n \le 37$, every branch of the finite domains in Lemma 35 is excluded by the preliminary results and the bounds proved in Lemmas 1–6, 22–29 and 30–41. The initial conductor exclusion in Section 11.1 closes $n = 13$ directly, and the recorded computation supplies the exclusions for $14 \le n \le 37$. The implementation details in Appendix C preserve these terminal exclusions. A strict violation would supply its Apéry profile, selected order ideal and primitive values in one of these finite domains, and would survive every necessary test. This contradicts the initial conductor exclusion or the closed finite cover, whose soundness is given by Proposition 42. □

11.5 Relation to the general verification theorem

The traversal just reported operates on profiles, selected monomials and primitive boxes. It is distinct from $\textsc{Primary}(C, u, v)$ and the forest encoding of Sections 4–5. The present paper proves the latter specifications and their resource bounds, but supplies no new implementation or execution of them. In particular, the recorded residual exclusions are not represented as an accepted general forest certificate.

For this application, a future accepted `VERIFIED(200,13,37)` certificate would replace the external conductor input on exactly the needed left-cardinality interval, while the retained residual cover would handle the rest. The conditional composition follows by separating $c \le 200$ from $c > 200$. The stronger all-cardinality statement through conductor 200 would require its full scope, for example `VERIFIED(200,1,100)`. Alternatively the scalar cutoff permits a single general traversal through $H_{37} = 1292$. None of these new certificates is asserted to exist with verified nonnegative outputs on the basis of the historical computation alone; the existing theorem continues to use its stated external input.

A A graph-theoretic reduction

Proposition 44. *If $e, q \ge 4$ and $n \le 3q - 1$, then $W(S) \ge 0$.*

Proof. Let $X = A \setminus \{0\}$. For $x \in X$ define its graph depth $d_G(x)$ by $x + d_G(x)m \in [c, c + m)$. If $x \in A_i$, then $d_G(x) = q - i$. Proposition 2.9 and Corollary 2.10 of [5] give

$$\tau(X) := \sum_{x \in X} d_G(x) = n - q, \qquad m = e + \delta, \qquad W = e\tau(X) - q\delta + \rho. \tag{86}$$

Thus $n \le 3q - 1$ is equivalent to $\tau(X) \le 2q - 1$.

The graph in that paper has edges $\{x, y\}$ whenever $x, y \in X$ and $x + y \in X$, with loops allowed and only incident elements retained as vertices. Write V for this vertex set and $b = |V|$; this avoids the source's use of n for a quantity different from our $|L|$. Write k for the largest *number of vertices touched* by a matching (a loop touches one vertex), and ν_G (the source's ν) for the largest number of vertices touched by normal edges in a k-vertex matching. An edge is normal when $d_G(x) + d_G(y) \ge q$ and weak when that sum is $q - 1$. The source's Theorem 3.24 is

$$\tau(X) \ge \frac{k(q-1) + \nu_G}{2} + (b - k).$$

Under $q \ge 4$ and $\tau(X) \le 2q - 1$, Proposition 5.3 consequently gives $k \le 4$: $k \ge 5$ would give $\frac{5}{2}(q-1) \le 2q-1$ and hence $q \le 3$. The complete case argument in its §§5.2–5.4 proves $W \ge 0$ under these hypotheses and $e \ge 4$. More explicitly, §5.2 treats $k = 0, 1, 2$; Lemma 5.4, Proposition 5.5, and the remaining loop-count cases of §5.3 treat $k = 3$. For $k = 4$, Lemma 5.6 gives $b \in \{4, 5\}$ and $\nu_G \le 2$; $b = 5$ forces $\nu_G = 0$ and $\tau(X) = 2q - 1$. Propositions 5.7–5.9 and the ensuing loop-count cases, with the correction below, treat $b = 5$, while Propositions 5.10–5.12, also corrected below, treat $b = 4$, according as $|V \cap D|$ is 0, 1, or at least 2. The supporting edge-weight, weak-edge and graph statements used there (Propositions 3.11–3.12, Corollary 3.21, Theorem 3.24, and the cited statements of §4) concern every such semigroup graph and have no embedding ratio hypothesis.

For clarity, the imported case split and its hypotheses are as follows. Here the source's graph-vertex count has been renamed b; in particular none of the rows replaces our $n = |L|$ by the source's n.

matching size k	graph subcase	cited proof portion
$0, 1, 2$	all b allowed by the depth bound	[5, §5.2]
3	all loop-count cases	[5, Lemma 5.4, Proposition 5.5, §5.3]
4	$b = 5$	[5, Lemma 5.6, Propositions 5.7–5.9]
4	$b = 4$, $\lvert V \cap D \rvert = 0, 1, \geq 2$	[5, Propositions 5.10–5.12], corrected below

Each row is used under $q \geq 4$, $e \geq 4$ and $\tau(X) \leq 2q - 1$. **Graph conventions.** We first record two corrections to the argument in [5] in arXiv:1909.03699v1. Proposition 5.8's ten-edge graph is K_5, as established by Proposition 5.7, rather than the printed LK_5 (which has fifteen edges and touches five vertices in a matching). The regularity argument remains valid for K_5. In §5.2 a loopless star on b vertices has $b - 1$ edges; use the conservative $\delta \leq b$ in the printed bound. The displayed lower bound then expands to

$$q + (b + 1)(b - 3) + (b + 1)\nu_G/2 + \rho,$$

which is at least the source's following expression $q + b(b - 3) + b\nu_G/2 + \rho$ in that case $b \geq 5$. These corrections do not introduce a ratio premise. Throughout, the source's depth symbol is d_G here, never the decomposable count δ.

Five vertices and one loop. The classification in [5, §5.4.1] needs a replacement. On five vertices, the graph consisting of a loop at x, the four edges xa, xb, xc, xd, and the triangle ab, ac, bc has eight edges and $k = 4$. It is not the asserted join of a loop with a claw. The required semigroup subcase is nevertheless excluded directly, as follows.

Suppose $b = 5$, $k = 4$, there is exactly one loop, $\delta \geq 8$, and $\lvert V \cap D \rvert \geq 2$. Delete the sole loopy vertex x. The remaining true edges are pairwise intersecting: two disjoint edges, together with the loop at x, would touch five vertices. A pairwise intersecting family of edges on four vertices is a star or a triangle, and has at most three edges. Therefore $\lvert E \rvert \leq 1 + 4 + 3 = 8$. Surjectivity of the edge-weight map gives $\delta \leq \lvert E \rvert$, so $\delta = \lvert E \rvert = 8$ and that map is injective. Every decomposable incident vertex is left, by Lemma 1. Lemma 4's collision argument shows that any such vertex with at least two distinct neighbors would give two edges with the same weight. Hence every vertex of $V \cap D$ has degree one, counting a loop as one neighbor; this also follows from [5, Proposition 4.17]. But deleting two degree-one vertices removes at most two edges, and the remaining three vertices have at most three true edges and the single loop. Thus $\lvert E \rvert \leq 2 + 4 = 6$, contradicting $\lvert E \rvert = 8$. This proves the required exclusion without the disputed classification.

Four primitive vertices. In the primitive-vertex case of [5, Proposition 5.10], the assertion that $\nu_G = 2$ permits at most three normal edges is too strong. The following replacement closes both affected subcases, without an embedding-ratio assumption. Here $b = k = 4$, $V \subseteq P$, and $\nu_G = 2$, so $e \geq 5$ and Theorem 3.24 together with $\tau(X) \leq 2q - 1$ gives $\tau(X) = 2q - 1$.

First suppose $\delta = 10$. There are at most ten edges on four vertices, including loops, and the edge-weight map is onto D. Hence the graph is LK_4 and its ten edge weights are distinct. Every matching of normal edges extends to a matching touching all four vertices, by adding the loops at untouched vertices. Thus no matching of normal edges touches more than two vertices. The true normal edges are consequently pairwise intersecting, and are a star or a triangle. A triangle permits no normal loop, since a loop and its opposite edge would touch three vertices. In a star with at least two true edges, only the center can support a normal loop, for the same reason. These cases have at most three and four normal edges, respectively. With one true edge there are at most its two endpoint loops, and with no true edge there are at most two loops. Therefore there are at most four normal edges and at least six weak edges. Their distinct weights and Corollary 3.21 imply $\rho \geq 6$. Consequently

$$W = e\tau(X) - q\delta + \rho \geq 5(2q - 1) - 10q + \rho \geq 1.$$

If instead $\delta = 9$, no normal-edge count or injectivity assumption is needed. We have

$$W \geq 5(2q-1) - 9q + \rho = q - 5 + \rho.$$

This is nonnegative for $q \geq 5$. When $q = 4$, a matching touching four vertices must contain a weak edge, because $\nu_G = 2 < 4$. Its weight supplies one element of the weak-weight image, so Corollary 3.21 gives $\rho \geq 1$ and again $W \geq 0$. For completeness, the neighboring weak-edge counts must count distinct weights. For any subset F of the edge set E, surjectivity of the weight map gives

$$|\operatorname{wt}(F)| \geq |F| - (|E| - \delta).$$

Indeed, the total number of repetitions over all weight fibers is $|E| - \delta$, and restricting to F cannot increase that number. In the following two cases, an edge is *active* if it belongs to a matching touching all four vertices, and *passive* otherwise. When $\delta = 8$, there are at most two edges missing from LK_4. The following explicit completion witnesses make the edge count precise; write aa for a loop and ab for a true edge on vertices a, b, c, d.

present edge	ways to complete to four touched vertices	condition for passivity
aa	$\{bb, cd\}, \{cc, bd\}, \{dd, bc\}$	at least three missing edges
ab	$\{cd\}$ or $\{cc, dd\}$	cd and at least one of cc, dd missing

The three displayed completions for aa have disjoint edge sets, so at most two missing edges leave one intact. Two distinct true edges cannot both meet the second row's passivity condition with only two edges missing: each has a distinct complementary edge, and the remaining missing loop can belong to at most one required complementary pair. Thus at most one present edge is passive, and at least $|E| - 1$ are active. If $\nu_G = 0$, they are weak, and the displayed fiber bound gives $\rho \geq 7$, which more than suffices in $W \geq 2q - 10 + \rho$. When $\delta = 9$, at most one edge is missing, so all edges are active. If $\nu_G = 1$, no true edge is normal, and at most one loop is normal: two normal loops can be completed on the other two vertices by their edge or their two loops, since at most one edge is missing. Thus at least $|E| - 1$ edges are weak, giving $\rho \geq 8$ by the same fiber bound. This also validates that neighboring subcase without treating weak edges as automatically having different weights. The other subcases retain their stated conclusions. This completes the case argument.

□

B The profile-exclusion procedure for the recorded application

Floors and ceilings retain their mathematical meaning for negative arguments. This appendix specifies the finite search for $14 \leq n \leq 37$. The bounds are proved in the preceding sections. Set $H = 200$. All integer intervals are inclusive, and all ceilings, floors and comparisons use exact signed arithmetic.

B.1 Control flow and matching

The operations in the following pseudocode are defined by the formulas and loops below. A branch is closed only by a necessary contradiction. An unresolved branch prevents the conclusion `CLOSED`.

```
Cover(n):                                  # 14 <= n <= 37
  exclude e <= 3 by the cited theorem
  for q in 4,...,floor(n/3):
    for alpha in all weighted profiles sum((q-i)*alpha[i])=n-q:
```

```
    if early_empty(alpha): continue
    if raw_conductor_endpoint(alpha) <= H: continue
    for kappa in the complete rectangle 0 <= kappa[i] <= alpha[i]:
      require kappa[1] = 0
      beta = alpha - kappa
      state = RefineProfile(alpha, kappa, beta)
      if state is REJECTED: continue
      for h in AllGlobalAllocations(state):
        refined = RefineAllocation(state, h)
        if refined is REJECTED: continue
        for (e, delta, rho) in AllIntegerTuples(refined):
          m = e + delta; c = q*m - rho
          box = PrimitiveBox(beta, m, rho)
          if box is empty: continue
          if RankedCapsReject(refined, h, box, m, rho): continue
          for labeling in AllLeftLabelings(h, box):
            # quota or propagation contradictions prune locally
            if Pack(labeling, box, e, delta, m, rho) is UNRESOLVED:
              return UNRESOLVED
return CLOSED                          # every branch contradicted
```

```
Pack(labeling, box, e, delta, m, rho):
  box = PropagateAllConstraints(labeling, box)
  if box is empty: return CLOSED
  top = AllTopCandidates(labeling, box)
  T = IntegerMatching(top)
  R = ResidueMatching(zero, primitives, selected_left, top)
  if min(K + T, R - 1 - s) < delta: return CLOSED
  if every coordinate of box is a singleton: return UNRESOLVED
  i = first coordinate having largest width U[i]-L[i]
  z = floor((L[i]+U[i])/2)
  visit both boxes with coordinate i in [L[i],z] and [z+1,U[i]]
  return CLOSED iff both recursive calls return CLOSED
```

Each `Pack` call inherits the fixed depth q, the same global allocation h, and its associated counts K and s from the current refined state. Recursive box calls preserve these parameters; only the box and its consequent propagation and matching bounds change.

The preliminary results leave $4 \le q \le \lfloor n/3 \rfloor$. The loops below specify the weighted compositions, decomposable profiles, global allocations, integer tuples, and every admissible label. Stationary refinement states are retained. Both children of a split are visited, and `CLOSED` requires every terminal exclusion.

The integer matching operation sorts the nonempty integer intervals $I_u = [a_u, b_u]_{\mathbb{Z}}$ by increasing b_u and then a_u. Maintain a set of used integers. For each interval, assign its smallest unused integer if one exists, and otherwise skip it. The number of assigned intervals is $T_{\max}$ by Lemma 39. This is one operation on the union of all top candidate degrees.

For `ResidueMatching`, the right vertex set is $\{0, \dots, m-1\}$. For each left interval $[a, b]_{\mathbb{Z}}$, include exactly the residues r for which some integer in the interval is congruent to r modulo m. This set is all residues if $b - a + 1 \ge m$; otherwise it is obtained by reducing each integer in the interval. Start with the empty matching and search for an alternating path from an unmatched left vertex to an unmatched residue. Reverse matched and unmatched edges along each such path, and repeat until no augmenting path exists. The number of matched vertices is $R_{\max}$: if a larger matching existed, its symmetric difference with the current one would contain an augmenting path. Vertices from different degrees enter the same graph. Separate maxima for separate degrees cannot replace this joint maximum.

Thus the implementation needs only finite enumeration, exact integer arithmetic, interval propagation, and these two finite matching algorithms. The mathematical justification that an actual semigroup supplies witnesses surviving every necessary test is independent of the choice of programming language; it is Proposition 42.

B.2 Bounds used in the enumeration

Table 4. Bounds used in the enumeration.

Stage	Applicable estimates
Weighted profiles and eligible pairs	Complete weighted compositions and unordered pairs with loops: Lemmas 3, 35.
Macaulay growth and shadows	Positive-degree binomial growth, including zero and small dimensions: Lemma 22.

Stage	Applicable estimates
Factor patterns and split bounds	Repeated-factor patterns, every bipartition, occupied proper sums and universal shadow endpoints: Lemmas 23–26.
Ordinary and noncompressed caps	Universal numerical shadows, forced compressed pairs, conditioned patterns and applicable cubic/quartic minima: Lemmas 24–27, 30–32.
Profile refinement and conductor endpoints	Full-e endpoints; one global ρ allowance; alternative star/collision and depth-four bounds: Lemmas 4–6, 30–34.
Global allocations and selected shadows	One global h, complete bounded compositions, all selected-shadow degrees, exact left/top split and terminating refinements: Lemmas 28, 29, 30, 35.
Exact integer tuples	Every full-e, δ pair and every retained integer ρ, with the exact complementary exclusions in (91): Lemmas 5, 6, 35.
Primitive boxes and ranked caps	Simultaneous ranks and all-divisor interval tests retaining the same h: Lemmas 29, 36.
Labels, divisors and propagation	All degrees, exact quotas, zero-label meaning, every immediate divisor, and integer multiplicities: Lemmas 37, 38.
Integer and residue matching	Exact joint maxima across all degrees, residue wraparound, conservative omission of right primitives: Lemmas 39, 40.
Box splitting and closure	Both children, genuine unresolved leaves, complete scope and closed-cover implication: Lemma 41, Proposition 42.

B.3 Finite counting operators

Here is a constructive definition of every cap called by the traversal. For $s \geq 0$ and $j \geq 1$, obtain $a_j, \ldots, a_1$ by descending greedy subtraction: at position i choose the largest $a_i \geq i$ with $\binom{a_i}{i} \leq s$ if $s > 0$, subtract that binomial, and use the zero term $\binom{i-1}{i} = 0$ if $s = 0$. Put $G_j(s) = \sum_{i=1}^{j} \binom{a_i+1}{i+1}$ and $H_{d,t}(s) = G_{t-1} \circ \cdots \circ G_d(s)$, including $H_{d,d}(s) = s$. This is the convention of Lemma 22; every loop is finite.

For $2 \leq t \leq Q \leq q$, form the finite multiset-pattern set

$$\mathcal{P}(Q,t) = \left\{ l_1 \leq \cdots \leq l_t : \ 1 \leq l_i \leq Q - t + 1, \quad \max\{l_t + t - 1, \sum_i l_i - t + 1\} \leq Q \right\}. \tag{87}$$

The count of named monomials with pattern l is $v_\beta(l) = \prod_i \binom{\beta_i + r_i(l) - 1}{r_i(l)}$, with factor one if $r_i(l) = 0$ and factor zero if $r_i(l) > 0 = \beta_i$. Every pattern filter below first requires the all-bipartition recursion of Lemma 25: singleton i has $\mu(i) = i$ when $\beta_i > 0$; an unavailable proper piece makes the whole pattern unavailable; otherwise take the maximum over *all* nonempty bipartitions in (47), then round a proper nonsingleton to the first occupied κ-level not exceeding its indicated upper bound. Call the unrounded full-target lower bound $a(l)$ and its noncompressed version $b(l)$. A full target at level at most Q passes if its applicable lower bound is at most Q; for exact level Q additionally require $\sum_i l_i + t - 1 \geq Q$. Its formal cap is the sum of $v_\beta(l)$ over passing $l \in \mathcal{P}(Q,t)$.

Write $P_Q(z)$ and $U_Q(z)$ for the two pair sums in (9), with $z_i = 0$ outside $1 \leq i < Q$. For each proper degree d, define $a_d(Q)$ exactly as in Lemma 26, with sentinel Q when no proper occupied pattern exists. For $2 \leq d < t$, let

$$S_{Q,t,d}(\kappa) = \sum_{a_d(Q) \leq j \leq \min\{Q-t+d,\, Q - a_{t-d}(Q)+1\}} \kappa_j,$$

where the final $+1$ is deleted in the noncompressed case and an empty range sums to zero. Intersect the formal cap for degree t with $H_{d,t}(S_{Q,t,d})$ for every $d = 2, \dots, t-1$. For $t = 2$ use only the pair count. At $Q = 5, t = 3$, also intersect with $\binom{\beta_1+\beta_2+2}{3} + \kappa_3\beta_3$ if $\kappa_2 = 0$; for the noncompressed cap intersect with $\binom{\beta_1+2}{3} + (\kappa_2+\kappa_3)\beta_2 + \kappa_2\beta_3$. At $Q = 5, t = 4$, $\beta_1 = 1$, and $\kappa_2 + \kappa_3 \le 3$, intersect the noncompressed degree-four cap with 3. Summing these degree caps gives $G(Q, \beta, \kappa)$ in the ordinary case. Its noncompressed analogue $N(Q, \beta, \kappa)$ replaces its pair cap by

$$N_2 = U_Q(\beta) - \sum_{j=2}^{Q-1} f_j,$$
$$f_j = \begin{cases} \max\{0, \kappa_j - p_j^{\mathrm{nc}}\}, & \kappa_j > 0 \text{ and every exact-level} \\ & \text{length } 3, \dots, j \text{ formal cap vanishes,} \\ 0, & \text{otherwise,} \end{cases} \tag{88}$$

where p_j^{nc} is the exact-level noncompressed formal pair cap. Reject if $N_2 < 0$. These are Lemmas 26, 27, 31, and 32. The ordinary prefix cap used before this refinement may equivalently start with Lemma 24's $P_Q(\beta) + \sum_{t=3}^{Q} \min_d H_{d,t}(a_{Q,t,d})$; the fixed traversal uses the stronger G just specified.

For a remaining conductor at least $H + 1$, a current positive $M_{\max} \ge m$ adds one filter to each full-target formal count. Let $L = \sum_i l_i$ and $r(l) = \#\{i : l_i \ge 2\}$. Delete a pattern if

$$r(l) \le 1 \text{ and } L \ge Q + 1, \quad \text{or} \quad r(l) \ge 2 \text{ and } (L - Q - 1)M_{\max} \ge (r(l) - 1)(qM_{\max} - H - 1). \tag{89}$$

Equality in the second test deletes the pattern because the necessary inequality in Lemma 30 is strict. Denote the resulting ordinary or noncompressed formal cap by $\Phi_{Q,t}^{(\mathrm{nc})}(M_{\max}; \text{exact})$; an "exact" flag also imposes $L + t - 1 \ge Q$. In the conditioned ordinary prefix cap $G_H(Q)$, keep $P_Q(\beta)$ as the pair term, use these filtered formal caps for $t \ge 3$, and intersect them with all shadow caps above. At $Q = 5, t = 4$, $\beta_1 = 1$, $\kappa_2 + \kappa_3 \le 3$, add the marked-quartic cap 3 if both $2M_{\max} \ge 3(qM_{\max} - H - 1)$ and $M_{\max} \ge 2(qM_{\max} - H - 1)$; use 4 if only the first holds. The $Q = 5, t = 3$ cap under $\kappa_2 = 0$ remains in force.

For one *fixed* global allocation h, define the selected cap $C_{Q,t}^{(\mathrm{nc})}(h, M_{\max})$ as the minimum of the filtered formal cap and, when $Q = q$, the exact top split $\sum_{j<q} h[t, j] + \Phi_{q,t}^{(\mathrm{nc})}(M_{\max}; \text{exact})$. For each $2 \le d < t$, also intersect with

$$H_{d,t}\left(\sum_{a_d(Q) \le j \le \min\{Q-t+d, Q-a_{t-d}(Q)+\epsilon\}} h[d, j]\right), \qquad \epsilon = \begin{cases} 1, & \text{ordinary,} \\ 0, & \text{noncompressed.} \end{cases}$$

The degree-two cap has no shadow term. All these are upper caps; taking their minimum never adds two unsupported losses.

B.4 The domain and refinement loops

The following numbered procedure fixes both the loop order and each discard condition for the common reduction on $13 \le n \le 37$. The recorded runs cover $n = 14, \dots, 37$; the initial estimates already close $n = 13$, as calculated in Section 11.1.

1. First exclude full embedding dimensions $e \le 3$ by the cited small-embedding-dimension result. For each n, visit $q = 1, \dots, n$. Dispose of $q = n$ by Lemma 2, $q \le 3$ by the imported small-depth result, and $n \le 3q - 1$ by the corrected small-total-depth consequence proved in Proposition 44 (now $e, q \ge 4$). The residual depths are $4 \le q \le \lfloor n/3 \rfloor$. For each, enumerate all $\alpha \in \mathbb{Z}_{\ge 0}^{q-1}$

satisfying (68): initialize $R_{\mathrm{rem}} = n - q$ and recursively choose $\alpha_i \in [0, \lfloor R_{\mathrm{rem}}/(q-i) \rfloor]_{\mathbb{Z}}$ for $i = 1, \dots, q-2$, subtract $(q-i)\alpha_i$ from R_{rem}, and set $\alpha_{q-1} = R_{\mathrm{rem}}$ after the last subtraction. The early-empty input disposes of an α exactly when $\alpha_i = 0$ for every $1 \le i < \lceil q/2 \rceil$.

2. Compute $(M, U) = (P_q(\alpha), U_q(\alpha))$, $\sigma = \min\{qM, (q-1)M + U\}$, $E = \lfloor (\sigma - 1)/(n-q) \rfloor$, and $C = qE + \sigma$. If $C \le H$, use the inclusive conductor input to discard the entire α family. Otherwise enumerate every point of the complete κ rectangle $\kappa_1 = 0$ and $\kappa_i \in [0, \alpha_i]_{\mathbb{Z}}$ for $2 \le i < q$; set $\beta = \alpha - \kappa$, $K = \sum \kappa_i$, $s = \sum \beta_i$, and $e_{\min} = \max\{4, 1 + s\}$.

3. For every $2 \le Q < q$, reject a rectangle point if $\sum_{j \le Q} \kappa_j > G(Q, \beta, \kappa)$. Otherwise initialize

$$F = \min\{M - \kappa_2 \alpha_{q-1}, G(q, \beta, \kappa)\}.$$

   If some occupied κ_j has at least two eligible T-neighbors (equivalently $\sum_{i \le \min(q-1, q+1-j)} \alpha_i \ge 2$), further intersect F with $M - 1$. If $K > 0$ and $\kappa_j = 0$ for every $j < q - 1$, intersect it with $P_q(\beta) + \binom{\beta_1 + \beta_2 + 2}{3}$. Compute $N = N(q, \beta, \kappa)$ by (88) and reject if its pair term is negative.

   For $q = 4$, also apply the exact hypotheses and caps of Lemmas 33–34. If $\kappa_2 = 0$, let $P_0 = \binom{\alpha_1 + \alpha_2 + 1}{2}$. When also $\kappa_3 = \alpha_3$, reject if $P_0 < \kappa_3$ and otherwise set $V = P_0 - \kappa_3 + \binom{\alpha_1 + 1}{2} + \alpha_1 \alpha_2 + \binom{\alpha_1 + 2}{3}$. If $\kappa_2 = 0$, intersect F with $P_0 + G_2(\kappa_3) + (\alpha_1 + \alpha_2)(\alpha_3 - \kappa_3)$. If $\alpha_1 = 0$, put $h_0 = \min\{\alpha_2 \alpha_3, \alpha_2(\alpha_3 - \kappa_3) + G_2(\kappa_3)\}$, intersect F with $\binom{\alpha_2 + 1}{2} + h_0$, and put $J = 2\alpha_2(\alpha_2 + 1) - \kappa_3 + 3h_0$. Here P_0 is defined whenever $\kappa_2 = 0$; V exists only under its extra $\kappa_3 = \alpha_3$ hypothesis; J exists only under $\alpha_1 = 0$.

4. Define the endpoint operator for current caps by

$$\begin{aligned} \sigma = \min\{&qF, (q-1)F + U, (q-1)F + N, \sigma_{\mathrm{old}}, \\ &3F + V \text{ if defined}, J \text{ if defined}\}, \\ E = \left\lfloor \frac{\sigma - 1}{n - q} \right\rfloor, &\qquad C = qE + \sigma. \end{aligned} \tag{90}$$

   Omit σ_{old} only on the first evaluation; keep every other previously obtained cap. If $E < e_{\min}$ or $C \le H$, reject. Otherwise put $M_{\max} = E + F$, reject if a conditioned prefix $\sum_{j \le Q} \kappa_j > G_H(Q)$ for any $2 \le Q < q$, and replace F by $\min\{F, G_H(q)\}$. Re-evaluate (90) with the old σ included. Continue until (F, E, σ) is unchanged or a rejection occurs. The caps are nonincreasing integers, so this loop terminates; its stationary state is retained, not rejected.

5. Allocate the one global order ideal layer by layer. Initialize $h[d, j] = 0$ for $2 \le d \le q$, $1 \le j < q$. At level $j = 2, \dots, q-1$, enumerate *all* tuples $(h[2, j], \dots, h[j, j])$ of nonnegative integers summing to κ_j, with coordinate cap $h[d, j] \le \Phi_{j,d}(E + F; \text{exact})$; coordinates $d > j$ remain zero. After each level, reject a partial allocation if $\sum_{i \le j} h[d, i] > C_{j,d}(h, E + F)$ for some $2 \le d \le j$. All remaining completed compositions are visited. The same h is passed to every subsequent predicate.

6. For a completed h, repeat a second fixed-point loop while $E \ge e_{\min}$ and $C > H$. With $M_{\max} = E + F$, reject if any $\sum_{j \le Q} h[t, j] > C_{Q,t}(h, M_{\max})$ for $2 \le t \le Q < q$. Replace

$$\begin{aligned} F &\leftarrow \min\left\{F, \sum_{t=2}^{q} C_{q,t}(h, M_{\max})\right\}, \\ N &\leftarrow \min\left\{N, \min\{N_2, C^{\mathrm{nc}}_{q,2}(h, M_{\max})\} + \sum_{t=3}^{q} C^{\mathrm{nc}}_{q,t}(h, M_{\max})\right\}. \end{aligned}$$

Recompute σ, E, C using $\min\{qF, (q-1)F+U, (q-1)F+N, \sigma_{\text{old}}\}$. Stop at unchanged (F, N, E, σ), or reject if $E < e_{\min}$ or $C \le H$. This is a monotone fixed-point calculation using the one h; no different allocation is substituted for a different shadow.

7. For each retained allocation, visit every $e \in [e_{\min}, E]_{\mathbb{Z}}$ and every $\delta \in [K, F]_{\mathbb{Z}}$. Put $m = e + \delta$ and visit each integer ρ in

$$\begin{gathered}[\max\{0, q\delta - \sigma, \delta - N\}, \min\{m-1, qm - ne - 1, qm - H - 1\}]_{\mathbb{Z}}\,; \\ c = qm - \rho.\end{gathered} \tag{91}$$

The complement is accounted for by, respectively, $\rho < 0$, $q\delta - \rho > \sigma$, $\delta > \rho + N$, $\rho \ge m$, $c \le ne$, or $c \le H$. The strict inequalities explain the subtraction of one at both right endpoints. No full embedding dimension is inferred from s: every e in this interval is visited.

B.5 Exact tuples, labels and box trees

For each tuple (e, δ, m, ρ, c) from (91), name the β_i primitive variables at level i in increasing value order. Flatten these names in increasing level and within-level index order, and use the simultaneous box (71); an empty coordinate rejects the tuple. The following recursive test is the ranked cap used at this exact tuple. For each degree-t multiset u of named variables, enumerate every nonempty bipartition $v + w = u$ and every proper divisor recursively. For a singleton $v = (i)$, its possible occupied level is its assigned primitive level. For a nonsingleton proper v, first set $a(v) = \lfloor (\sum_{i \in v} L_i + \rho)/m \rfloor$ and intersect it with all split lower bounds $\mu(v_1) + \mu(v_2) - 1$, $\mu(v_1) + 1$, $\mu(v_2) + 1$. Return the smallest j with $\kappa_j > 0$ satisfying

$$a(v) \le j \le \min\{q - 1, \sum_{i \in v} \lambda_i + |v| - 1, \lfloor (\sum_{i \in v} U_i + \rho)/m \rfloor\}$$

and whose inclusive level interval $[jm - \rho, (j+1)m - \rho - 1]_{\mathbb{Z}}$ meets $[\sum L_i, \sum U_i]_{\mathbb{Z}}$; if no such j exists, mark v unavailable. At a full target, use the same split lower bounds, increased to $\mu(v_1) + \mu(v_2)$ for a noncompressed target, and require the resulting lower bound at most Q, the pattern bounds in (87), and overlap with level at most Q. For an exact-level target also require the pattern's exact-level condition and upper box sum at least $Qm - \rho$. Count every named multiset passing these tests. Denote this integer by $B^{(\text{nc})}_{Q,t}$; it is zero if the initial box is empty. For each $Q < q$, reject if $\sum_{j \le Q} h[t, j] > \min\{C_{Q,t}, B_{Q,t}\}$ for some $2 \le t \le Q$. At $Q = q$ set

$$F' = \min\{F, \sum_{t=2}^{q} \min(C_{q,t}, B_{q,t})\}, \qquad N' = \min\{N, \sum_{t=2}^{q} \min(C^{\text{nc}}_{q,t}, B^{\text{nc}}_{q,t})\}.$$

Here every selected cap $C^{(\text{nc})}_{Q,t}$ retains the stationary $M_{\max} = E + F$ from the preceding allocation refinement. The ranked cap $B^{(\text{nc})}_{Q,t}$ uses the exact tuple m, ρ and its primitive box. Reject if $\delta > F'$ or $\delta > \rho + N'$; otherwise begin labels. These ranked counts test all proper divisors against one simultaneous box. They are upper bounds, even when the box includes inconsistent vectors.

Let $r = \max(\{1\} \cup \{d : \sum_j h[d, j] > 0\})$ and $t_{\max} = \min(q, r + 1)$. Enumerate all multisets of the s named variables in every degree $d = 2, \dots, r$, in increasing degree and lexicographic variable-index order. At entry to each degree, record, for every multiset u, every positive label $j \in [2, q-1]_{\mathbb{Z}}$ for which $h[d, j]$ has remaining quota, u passes the exact-level ranked test for j, and, if $d > 2$, every *distinct* immediate divisor was positively labeled at a level below j. These recorded possibilities may be a superset after later box tightening. Visit a positive label only while its quota is positive; then

decrement the quota and add the layer constraint $jm - \rho \le \sum_i u_i x_i \le (j+1)m - \rho - 1$. Visit label zero exactly when the total quota is smaller than the number of unprocessed degree-d multisets. Zero means "not selected left"; it does not exclude selection at the top level. At each partial node, reject if remaining total quota exceeds remaining multisets or some label quota exceeds the number of remaining recorded eligible multisets. At the end of a degree every quota must be zero before entering the next degree.

After each positive label and after each box split, propagate every stored layer constraint in repeated passes. For an exponent vector u, $a = jm - \rho$, $b = (j+1)m - \rho - 1$, and each $u_i > 0$, set

$$L_i \leftarrow \max\left\{L_i, \left\lceil \frac{a - \sum_{k \ne i} u_k U_k}{u_i} \right\rceil\right\},$$
$$U_i \leftarrow \min\left\{U_i, \left\lfloor \frac{b - \sum_{k \ne i} u_k L_k}{u_i} \right\rfloor\right\}.$$

For adjacent variables of the same primitive level, also set $L_{i+1} \leftarrow \max(L_{i+1}, L_i + 1)$ and $U_i \leftarrow \min(U_i, U_{i+1} - 1)$. Iterate until no endpoint changes or some $L_i > U_i$; only the latter is a contradiction. Repeated factors use their exponent u_i in the denominator, including when the numerator is negative.

For a completed labeling and current box, build one joint top family. For every multiset u of degrees $2, \dots, t_{\max}$, include it if it is not labeled left, each distinct immediate divisor is labeled left when $|u| \ge 3$, it passes the ranked exact top-level test, and its interval

$$I_u = [\sum_i u_i L_i, \sum_i u_i U_i]_{\mathbb{Z}} \cap [qm - \rho, (q+1)m - \rho - 1]_{\mathbb{Z}}$$

is nonempty. Compute $T_{\max}$ as the *maximum* matching of these intervals to top-layer integers. An exact algorithm sorts intervals by increasing upper endpoint, then lower endpoint, and assigns the smallest unused integer in each interval, skipping an interval when no such integer exists. If $K + T_{\max} < \delta$, reject. Otherwise build one residue graph with left vertices for $\{0\}$, all s primitive intervals $[L_i, U_i]_{\mathbb{Z}}$, all K positively labeled left intervals $[\sum u_i L_i, \sum u_i U_i]_{\mathbb{Z}} \cap [jm - \rho, (j+1)m - \rho - 1]_{\mathbb{Z}}$, and every optional I_u. Connect a vertex to precisely the residues modulo m attained by its inclusive interval; if it has at least m points, this is every residue. Compute the exact unrestricted maximum $R_{\max}$ by augmenting paths. Right primitives are deliberately absent from this necessary matching. Reject if $\min\{K + T_{\max}, R_{\max} - 1 - s\} < \delta$. The two maxima must be computed over all degrees jointly; two separate degree-wise maxima do not give this test.

If neither test rejects and every box interval is a singleton, mark an *unresolved* leaf. Otherwise choose the smallest coordinate among those of maximum width $U_i - L_i$, let $z = \lfloor (L_i + U_i)/2 \rfloor$, and visit *both* children obtained from $[L_i, z]_{\mathbb{Z}}$ and $[z+1, U_i]_{\mathbb{Z}}$, respectively. Propagate and recompute both joint maxima in each child. A requested node limit also yields an unresolved leaf rather than an exclusion. The fixed production invocation used no such limit. The tree is closed only if every required label choice and both children of every split terminate in explicit necessary contradictions. For each n, the recorded profile, allocation, tuple and labeling trees contain every required child and have no unresolved terminal case. Closure is checked at each leaf.

The computation described in Section 11 closes every branch for $14 \le n \le 37$. Profile refinement suffices for $14 \le n \le 32$; the remaining five cases also use allocations, tuples, labels and boxes. Appendix C explains two details of the implementation. Proposition 42 proves that complete exclusion of these finite domains establishes Wilf's inequality.

C Implementation details

Two details relate the implementation to the enumeration rules of Appendix B. They concern an empty primitive interval and four selected-shadow caps.

C.1 An empty primitive box at $n = 37$

Proposition 45 (An empty primitive interval)**.** *For $n = 37$, the tuple $q = 5$, $\beta = (1, 4, 0, 0)$ and $(e, \delta, m, \rho, c) = (6, 50, 56, 55, 225)$ has no primitive value vector satisfying Lemma 36.*

Proof. For this tuple,

$$q = 5, \qquad \beta = (1, 4, 0, 0), \qquad (e, \delta, m, \rho, c) = (6, 50, 56, 55, 225).$$

For the sole level-one primitive, (71) gives

$$L = \max\{m + 1, m - \rho\} = 57,$$
$$U = 2m - \rho - \beta_1 = 112 - 55 - 1 = 56.$$

Thus its inclusive interval is $[57, 56]_{\mathbb{Z}} = \varnothing$. The tuple is therefore excluded.

In the implementation this interval was passed to ranked counting, which returns zero. Since $h[2, 3] = 1$, the selected-prefix test also excludes the same tuple. The recorded first reason is the selected-prefix contradiction, whereas the printed procedure excludes the empty box earlier. □

C.2 Stationary and exact multiplicity in four raw caps

Lemma 30 permits any valid upper bound $M_{\max} \geq m$. The refinement carries $M_{\max} = E + F$ into selected-shadow caps. At some exact-tuple calls the implementation used m instead. This is also a valid upper bound and can give a sharper *raw* selected cap. Table 5 gives the four reached ordinary caps whose raw values differ between the two choices. The displayed cap is for degree t and target through Q, before its intersection with the other bounds. Write $C(M_{\max})$ and $C(m)$ for its two raw values.

Table 5. The four reached raw selected-cap differences.

n	tuple IDs	Q	t	$M_{\max}$	m	$C(M_{\max})$	$C(m)$
33	144, 145	6	3	37	36	16	13
37	35, 36	4	3	58	57	35	34

For $n = 33$, comparison of 1 992 reached cap calls found precisely the two listed differences. Other minima give $F' = 28$, $N' = 16$ and rejection at both tuples under either parameter. For $n = 37$, comparison of 26 846 selected-cap calls over 1 469 tuples found precisely the two listed raw differences and no differences in final F', N' or tuple decisions. Thus the changes are masked by the other caps in the reached states. At $n = 37$ the first-reason discrepancy is the separate empty-box case above.

The two choices therefore give the same final bounds and exclusions at these four calls. Together with the empty-interval calculation above, this justifies the terminal exclusions used in the covering argument.

Computational companion and execution records. The Parallel Attack computational companion, version `companions-v1`, supplies the checking programs, case manifests and instructions for replay. The 96 compressed mathematical execution journals are supplied separately as `Wilf_execution_companions_v1.tar` in the companion release. The companion README gives the archive checksum, download location and commands for integrity checks, validation of recorded cases, and fresh re-execution. The repository is private at this stage, so access requires repository permission.

The mathematical journals and the 24 per-cardinality manifests retain the recorded computation for $n = 14, \dots, 37$. Archive integrity identifies the supplied bytes; it does not validate the exclusion arguments. Recorded-case validation and re-execution have the distinct scopes stated in Section 11.3. The proof for $n = 13$ and the cited conductor check for $c \le 200$ remain separate inputs. This distribution does not claim a new full-range enumeration or execution of the newer general algorithms.

D Resource bounds for the residual profile domains

This appendix bounds one explicit finite enumeration of the mathematical domains in Sections 2–9. It does not claim a source-level runtime analysis of the historical implementation. Fix $2 \le q < n$, write $\nu = n - q$, and assume that each combinatorial prefix is visited once. Refinement stops when no integer cap changes; propagation stops at stability or emptiness; primitive boxes use midpoint splits. A surviving singleton may be returned as unresolved.

D.1 Profiles, allocations and scalar tuples

Proposition 46. *The number of profiles satisfying $\sum_{i=1}^{q-1}(q-i)\alpha_i = \nu$ is bounded by each of*

$$\prod_{w=1}^{q-1}(\lfloor \nu/w \rfloor + 1), \qquad \binom{\nu+q-2}{q-2}, \qquad 2^{\nu-1}.$$

For a fixed profile, at most $2^{t_L} \le 2^{\nu}$ decomposable profiles are possible. For a fixed decomposable profile κ, the number of degree allocations before shadow tests is

$$\mathcal{H}(\kappa) = \prod_{j=2}^{q-1} \binom{\kappa_j + j - 2}{j-2} \le (q-2)^K \quad (q \ge 3), \qquad K = \sum_j \kappa_j.$$

For $q = 2$ the allocation count is one.

Proof. Each profile is a partition of ν into parts at most $q-1$. Independent coordinate bounds give the product. Choose the $q-2$ coordinates of weights at least two; their sum is at most ν, and the weight-one coordinate is then determined. Stars and bars give the second bound. Sorting the parts embeds partitions into compositions, of which there are $2^{\nu-1}$. An enumerator assigning coordinates in decreasing weight has at most q times as many prefixes as full solutions: each prefix that leaves a nonnegative remainder has a weight-one completion, and every such path has at most q nodes.

Since $\kappa_1 = 0$ and $0 \le \kappa_i \le \alpha_i$, the rectangle has $\prod_{i=2}^{q-1}(\alpha_i + 1) \le 2^{t_L}$ points, using $a + 1 \le 2^a$. At level j, its κ_j selected decomposable values are distributed among degrees $2, \dots, j$. Stars and bars give the displayed product. A composition into $j-1$ boxes is the image of a labeled assignment to those boxes, so its number is at most $(j-1)^{\kappa_j} \le (q-2)^{\kappa_j}$. □

All constraints on a branch use the same allocation; the count does not permit replacing it by unrelated allocations for different tests. Let F be the initial pair cap. Then $F \le t_L(t_L+1)/2 \le \nu(\nu+1)/2$. From $\sigma \le qF$ and the strict endpoint in Lemma 6, one has

$$e \le E \le q(\nu+1)/2, \qquad m = e+\delta \le E+F \le n(\nu+1)/2.$$

Ignoring further restrictions, the number of triples (e,δ,ρ) with $1 \le e \le E$, $0 \le \delta \le F$, $0 \le \rho < e+\delta$ is at most $E(F+1)(E+F) = O(n^6)$, and $c = qm - \rho = O(n^3)$ in this domain. If integer caps only decrease and refinement stops at equality, the number of strict decreases is at most the sum of their initial ranges. Domain counts alone would not bound an algorithm that redundantly refines forever.

D.2 Monomials, labels and boxes

The number of exponent vectors in s variables of degrees 2 through $d_{\max}$ is

$$M(s, d_{\max}) = \binom{s + d_{\max}}{d_{\max}} - 1 - s.$$

This follows by adding a slack coordinate to count degrees at most $d_{\max}$, then removing degrees zero and one. It counts monomials, with repetitions, rather than ordered words. The selected degrees used here are at most q, so $M(s,q) \le 2^{s+q} \le 2^n$. Smaller primitive width and degree yield smaller domains before any numerical or divisor test.

If M_L left monomials are labeled in a fixed order, each has zero or one of at most $q-1$ positive level labels. At most K positions may have a positive label. Including later-rejected prefixes, their number is

$$(M_L+1) \sum_{i=0}^{\min(K,M_L)} \binom{M_L}{i} (q-1)^i. \tag{92}$$

Indeed, at any depth choose the positive positions and their labels; there are at most M_L+1 depths, each bounded by the displayed full position count. Since $M_L \le 2^n$ and $K \le n$, this is $2^{O(n^2)}$. Immediate-divisor tests and quotas can only reduce the domain; admissible labels need not represent actual semigroups.

For a nonempty initial primitive box with inclusive widths $w_1, \dots, w_s$, put $P_{\mathrm{box}} = \prod_i w_i$, with the empty product equal to one. Midpoint splitting followed by shrinking propagation gives at most P_{box} terminal leaves and $2P_{\mathrm{box}} - 1$ nodes. To see this, associate to each leaf a point in its nonempty domain before its final propagation. Different leaves lie in disjoint split domains. Every split has two children, so the node count follows. A box initially empty is a separate one-node case. Splitting coordinate i at most $\lceil \log_2 w_i \rceil$ times makes it a singleton, and propagation can only help. Thus the path depth is at most $\sum_i \lceil \log_2 w_i \rceil$. Since $w_i \le m = O(n^2)$ and $s \le n$, $P_{\mathrm{box}} = n^{O(n)}$. Arbitrary splits would give only the weaker depth bound $\sum_i (w_i - 1)$.

D.3 Work within a box

Let R_{lin} be the number of linear monomial constraints and let b bound the bit lengths of coefficients, intermediate values and addresses. A direct pass recomputing each other-coordinate sum for each row and coordinate costs $O((R_{\mathrm{lin}} s^2 + s + 1) b^2)$, using exact signed floor/ceiling division. Every strict nonempty change reduces $\sum_i (U_i - L_i)$ by at least one. There are at most $1 + \sum_i (w_i - 1)$ passes before stability, and at most one further emptiness-detecting pass. This charges propagation explicitly.

Let T_{top} be the number of nonempty top intervals. The exact integer matching rule of Lemma 39 is implemented by sorting upper endpoints and keeping a sorted list of already assigned values. To find the least available value in an interval, scan that used-value list, advancing past occupied values. There are at most T_{top} such values, independent of the interval's numerical length. This costs $O((T_{\mathrm{top}}^2 + T_{\mathrm{top}} \log(T_{\mathrm{top}} + 1))b)$ time and $O(T_{\mathrm{top}} b)$ bits. It computes the proved maximum, not merely an arbitrary feasible matching.

For the residue graph, let $A_L = 1 + s + K + T_{\mathrm{top}}$ be its left vertex count and E_R its explicitly materialized edge count. An interval of length a contributes $\min(m, a)$ residues, including wraparound, so $E_R \le A_L m$. Construction costs $O(A_L b^2 + E_R b)$. Inserting left vertices one at a time and exhaustively searching alternating paths takes

$$O(A_L(A_L + m + E_R)b) \quad \text{time}, \qquad O((A_L + m + E_R)b) \quad \text{bits}.$$

A successful augmenting path increases the maximum by one; if exhaustive search fails, the symmetric difference with a hypothetically larger matching would contain such a path, a contradiction. Thus exact matching is obtained. A short interval description does not eliminate the cost of its materialized graph.

There are at most $M = M(s, q)$ named monomials. Store exponent vectors in a comparison dictionary, and compute intervals by their s-term dot products. Each vector has at most s immediate divisors. Comparing vectors of s coordinates in a balanced search tree gives the safe bound $O(Ms^2 \log(M+2)b^2)$ per complete rebuilding and $O(Msb)$ bits. Arithmetic values here need only $O(\log(n+2))$ bits, but addresses into up to 2^n records need $O(n)$ bits; $b = O(n+1)$ covers both.

The last dictionary estimate covers intervals and immediate-divisor checks. Ranked caps also require every proper divisor and nonempty bipartition, as in Lemmas 25 and 36. For an exponent vector $\mathbf{u}$ of degree at most q, the number of its divisors is $\prod_i (u_i + 1) \le 2^{\sum_i u_i} \le 2^q$. Enumerate these divisors explicitly and memoize the occupied-level recursion in increasing degree. This adds a factor at most 2^q times a polynomial in n to the per-vector work, and remains $2^{O(n)} \operatorname{poly}(n)$ per rebuilding.

The profile-pattern and shadow operators require their own accounting. There are at most $\binom{2q-1}{q} \le 4^q$ level-multiset patterns of degree at most q on $q-1$ levels; each has at most 2^q submultisets. Memoizing the all-bipartition recursion, with the polynomially many level, degree and shadow variants, again costs $2^{O(n)} \operatorname{poly}(n)$ per complete refinement pass. For a shadow input $a \le n$ and $1 \le d < t \le n$, monotonicity and the pure-binomial growth expansion give

$$H_{d,t}(a) \le H_{d,t}\left(\binom{n+d-1}{d}\right) = \binom{n+t-1}{t} \le 2^{2n}.$$

A Pascal table through row $2n$ therefore permits the binomial and greedy expansion arithmetic in polynomial bit work with $O(n)$-bit entries. Formal named-pattern counts also have $2^{O(n)}$ magnitude. The caps F, σ, E start polynomially bounded, and the noncompressed cap N may initially be intersected with F, since $|D \setminus \mathcal{C}| \le \delta$. Keeping these minima makes the number of strict cap decreases polynomial. These charges include the refinement operators, rather than treating all-bipartition or shadow calculations as unit-cost tests.

Combining these specified enumerators gives $2^{O(n^2)}$ time for a complete traversal of the relaxed residual domains at fixed n, plus the length of any retained certificate or other requested output. Profiles and rectangles contribute $2^{O(n)}$, allocations $n^{O(n)}$, scalar tuples a polynomial, labels (92), boxes $n^{O(n)}$, and per-box work at most $2^{O(n)}$ times a polynomial. The finitely many depths $q < n$ preserve the bound. This is an upper bound for the explicit scheme, with no average-case or

optimality claim. Successful exclusion of every leaf is a separate mathematical and computational condition, supplied for the recorded application only by its stated evidence in Section 11.

Acknowledgements

The arguments, implementations and exposition were developed with substantial assistance from OpenAI's Codex and GPT-family models.

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